Sufficient and Necessary Conditions for the Parameters of Conditional Entropies: Complete Proofs

Roberto Rubboli, Erkka Haapasalo, and Marco Tomamichel

This standalone companion gives the complete analytic proofs of the exact parameter ranges for the conditional-entropy candidates introduced in Ref. [1]. It follows the order and notation of the concise companion [3], but includes the real-power calculus, common discretization, signed-measure differentiation, two-block and three-block localization, the separate Shannon argument, tropical stationarity, and the lift from one-column shape inequalities to the original semiring and conditionally mixing channels. No global finiteness assumption is placed on the parameter measure. In the exceptional branches, the required one-sided moment finiteness is derived from monotonicity before the signed moment is formed.

CONTENTS

RELATION TO THE CONCISE COMPANION

Ref. [3] is the statement-and-main-ideas layer of this paper. The present paper proves the same numbered statements in the same order. Sections 1 and 2 keep the logical obstruction and the witness choices visible; Appendix A supplies the analytic infrastructure behind them. In particular, the finite counterexamples used here are the same Shannon, two-block, three-block, and binary-face witnesses recorded in the Lean correspondence note Ref. [2].

The proof dependencies can be read from left to right:

$$\begin{aligned}&\text{endpoint-aware finite-dimensional calculus} \longrightarrow \text{signed-measure differentiation}\\ &\quad\longrightarrow \text{two-/three-block and Shannon localization} \longrightarrow \text{finite curvature witness}\\ &\quad\longrightarrow \text{semiring/channel lift.}\end{aligned}$$

For sufficiency, a common discretization transfers the finite-support result to arbitrary measures. At fixed dimension, total variation controls integration against signed measures. For necessity, all potentially singular moments are first truncated away from $\alpha = 1$; localization extracts finite coefficients, and monotone convergence supplies one-sided finiteness only when the exceptional branch requires it. Exact finite-dimensional stationarity is needed only for the negative tropical branch. Finally, Section 1 A converts the one-column shape result into the original channel preorder.

1. SUFFICIENT CONDITIONS FOR THE PARAMETERS

In this section, we establish sufficient conditions on the parameters under which the homomorphisms and derivations obtained in Section 3 of Ref. [1] are monotone under conditional majorization. These conditions will be shown to be necessary in Section 2. The proofs below apply to arbitrary probability measures and do not impose a regularity assumption at the Shannon point $\alpha = 1$.

Let us first explain the structure of the argument. The homomorphisms in the temperate case are of the form

$$\Phi_{t,\tau}(P_{XY}) = \sum_{y\in\mathcal{Y}} P_Y(y) \exp\left(t \int_{[0,+\infty]} H_\alpha(P_{X|Y=y})\, \mathrm{d}\tau(\alpha) \right), \tag{1}$$

where $t \in \mathbb{R} \setminus \{0\}$ and τ is a probability measure on $[0, +\infty]$. Appendix A of Ref. [1] relates monotonicity under stochastic maps acting on the conditioning system to a concavity or convexity property of the corresponding function on a single unnormalised column. We give the complete reduction and its converse below. For $\vec{x} \in \mathbb{R}^d_{\geq 0} \setminus \{0\}$, consider

$$\vec{x} \longmapsto \Phi_{t,\tau}(\vec{x}) := \|\vec{x}\|_1 \exp\left(t \int_{[0,+\infty]} H_\alpha(\widehat{x})\, \mathrm{d}\tau(\alpha) \right), \qquad \widehat{x} := \frac{\vec{x}}{\|\vec{x}\|_1}. \tag{2}$$

We set $\Phi_{t,\tau}(0) := 0$. The required property is concavity when $t > 0$, corresponding to the codomain $\mathbb{R}_+$, and convexity when $t < 0$, corresponding to $\mathbb{R}_+^{\mathrm{op}}$.

The proof is organised as follows.

1. We first treat finitely supported measures. In this case, the function in (2) factorises into a monomial of power means. The admissible parameter range then follows from the curvature of the power means and of the outer monomial.

2. We extend the result to a general probability measure by a common discretisation of the parameter space. The word "common" is important: one and the same discrete measure is used at every vector appearing in a concavity or convexity inequality.

3. We verify monotonicity under doubly stochastic maps on the first system. This step is independent of the concavity or convexity analysis and follows from the Schur concavity of every Rényi entropy.

4. Finally, we obtain the tropical homomorphisms and the derivations. The tropical quantities arise as log-sum-exp limits, whereas the derivations are treated directly as perspectives of concave Rényi entropies.

For later use, we introduce the function

$$\omega(\alpha) := \frac{\alpha}{1-\alpha}, \qquad \alpha \in [0,+\infty] \setminus \{1\}, \qquad \omega(+\infty) := -1. \tag{3}$$

The value at $+\infty$ is the continuous limit of $\alpha/(1-\alpha)$ and is also the value compatible with the identity involving H_∞ used below. Throughout the paper, $\operatorname{supp}(\tau)$ denotes the closed support of a positive measure τ, and δ_a denotes the Dirac probability measure at a. If T is Borel measurable, its push-forward is defined by

$$(T_\# \tau)(B) := \tau\big(T^{-1}(B)\big) \tag{4}$$

for every Borel set B.

A. Reduction to one column and lift to conditionally mixing channels

We first make the connection between the one-column function and the original channel order explicit. Let P be a nonnegative matrix and write $p_y = P(\cdot, y)$ for its columns. A conditionally mixing channel is specified by doubly stochastic matrices S_k on the first register and nonnegative matrices D_k on the conditioning register, with

$$\sum_{k,y'} D_k(y,y') = 1 \qquad \text{for every } y. \tag{5}$$

Its output is

$$Q(x,y') = \sum_{k,y,z} S_k(x,z) P(z,y) D_k(y,y'), \tag{6}$$

and therefore its y'th column is

$$q_{y'} = \sum_{k,y} D_k(y,y') S_k p_y. \tag{7}$$

All summands are nonnegative. Since every S_k preserves mass,

$$\begin{aligned} \sum_{x,y'} Q(x,y') &= \sum_{k,y,y'} D_k(y,y') \sum_z P(z,y) \sum_x S_k(x,z) \\ &= \sum_y \left(\sum_{k,y'} D_k(y,y') \right) \sum_z P(z,y) = \sum_{z,y} P(z,y), \end{aligned} \tag{8}$$

so the formula preserves normalization when P is a probability matrix.

Let F be a positively homogeneous one-column functional. Concavity of F is equivalent to superadditivity on the nonnegative cone: applying concavity to $x/(\|x\|_1+\|z\|_1)$ and $z/(\|x\|_1+\|z\|_1)$, followed by positive homogeneity, gives $F(x+z) \geq F(x) + F(z)$. The same calculation shows that convexity is equivalent to subadditivity. Every Rényi entropy is Schur concave. Consequently, in the positive-temperature case,

$$F(S_k p_y) \geq F(p_y), \tag{9}$$

whereas in the negative-temperature case the negative exponential coefficient reverses this inequality:

$$F(S_k p_y) \leq F(p_y). \tag{10}$$

Using (7), superadditivity, and (5), the positive branch satisfies

$$\begin{aligned}\sum_{y'} F(q_{y'}) &\geq \sum_{y',k,y} D_k(y,y') F(S_k p_y) \\ &\geq \sum_{y',k,y} D_k(y,y') F(p_y) = \sum_{y} F(p_y).\end{aligned} \tag{11}$$

Convexity, subadditivity, and (10) give the reverse inequality in the negative branch. For derivations the relevant perspective is concave, so the first argument applies directly. The tropical inequalities follow either by the analogous max/min calculation or by the pointwise temperature limits proved below.

The converse shape implication uses only a two-column merge. Put positive multiples of x and z in two conditioning columns and apply the channel that merges them into the single column $\lambda x + (1-\lambda) z$. Channel monotonicity then gives exactly the concavity or convexity inequality for F. Finally, relabeling and adjoining zero rows or columns do not change any candidate. Embed matrices with different alphabets into a common zero-padded matrix, apply the fixed-alphabet channel inequality there, and remove the padding. This proves the original semiring/channel lift used throughout the paper without adding an axiom or an extra regularity hypothesis.

B. Sufficient conditions for the discrete case

This subsection proves Lemmas 1.1–1.3 of the concise companion.

The discrete proof is based on two elementary curvature results. We state them in the form needed here. Complete Hessian calculations are given in Appendix A.

Lemma 1.1. *Let $\alpha \in (0, +\infty]$. The function on the positive cone*

$$\vec{x} \longmapsto \|\vec{x}\|_\alpha := \left(\sum_{i=1}^{d} x_i^\alpha\right)^{1/\alpha}, \qquad \|\vec{x}\|_\infty := \max_i x_i, \tag{12}$$

is concave for $0 < \alpha < 1$ and convex for $1 \leq \alpha \leq +\infty$.

Lemma 1.2. *Let $N \in \mathbb{N}$ and let $\beta_0, \ldots, \beta_N \in \mathbb{R}$ satisfy*

$$\sum_{\ell=0}^{N} \beta_\ell = 1. \tag{13}$$

Then the function

$$(y_0, \dots, y_N) \longmapsto \prod_{\ell=0}^{N} y_\ell^{\beta_\ell} \tag{14}$$

on $\mathbb{R}_{>0}^{N+1}$ *is*

1. *concave if and only if* $\beta_\ell \geq 0$ *for every* ℓ*;*
2. *convex if and only if exactly one coefficient is positive and all the remaining coefficients are nonpositive.*

Moreover, the monomial is increasing in a coordinate with positive exponent and decreasing in a coordinate with negative exponent.

We now combine these two results. The statement includes the endpoint values $\alpha = 0, 1, +\infty$. Their treatment is explained explicitly in the proof.

Lemma 1.3. *Let* τ *be a probability measure supported on finitely many points of* $[0, +\infty]$*, and let* $t \in \mathbb{R} \setminus \{0\}$*. The function in* (2) *is*

1. *concave if* $t > 0$*,*
$$\operatorname{supp}(\tau) \subseteq [0,1), \qquad \int_{[0,1)} \frac{\alpha}{1-\alpha} \, \mathrm{d}\tau(\alpha) \leq \frac{1}{t}; \tag{15}$$
2. *convex if* $t < 0$ *and either*
   (a) $\operatorname{supp}(\tau) \subseteq [0,1]$*; or*
   (b) there exists $\alpha_* \in (1, +\infty]$ *such that*
$$\operatorname{supp}(\tau) \subseteq [0,1) \cup \{\alpha_*\}, \qquad \int_{[0,+\infty]\setminus\{1\}} \frac{\alpha}{1-\alpha} \, \mathrm{d}\tau(\alpha) \leq \frac{1}{t}. \tag{16}$$

Proof. We first assume that the support of τ does not contain 0 or 1. Write

$$\tau = \sum_{\ell=1}^{N} \tau_\ell \delta_{\alpha_\ell}, \qquad \tau_\ell > 0, \qquad \sum_{\ell=1}^{N} \tau_\ell = 1. \tag{17}$$

For every finite $\alpha \neq 1$, we have

$$\begin{aligned} H_\alpha(\widehat{x}) &= \frac{1}{1-\alpha} \log\left(\sum_i \left(\frac{x_i}{\|\vec{x}\|_1} \right)^{\alpha} \right) \\ &= \frac{1}{1-\alpha} \left(\log \sum_i x_i^\alpha - \alpha \log \|\vec{x}\|_1 \right) \\ &= \frac{\alpha}{1-\alpha} \left(\log \|\vec{x}\|_\alpha - \log \|\vec{x}\|_1 \right). \end{aligned} \tag{18}$$

The same identity holds at $\alpha = +\infty$ when $\|\vec{x}\|_\alpha$ is replaced by $\|\vec{x}\|_\infty$ and $\omega(+\infty) = -1$. Indeed,

$$H_\infty(\widehat{x}) = -\log \max_i \frac{x_i}{\|\vec{x}\|_1} = \log \|\vec{x}\|_1 - \log \|\vec{x}\|_\infty. \tag{19}$$

Define

$$\beta_\ell := t\,\omega(\alpha_\ell)\tau_\ell, \qquad \overline{\beta} := 1 - \sum_{\ell=1}^{N} \beta_\ell. \tag{20}$$

Substituting (18) into (2) gives the exact factorisation

$$\Phi_{t,\tau}(\vec{x}) = \|\vec{x}\|_1^{\overline{\beta}} \prod_{\ell=1}^{N} \|\vec{x}\|_{\alpha_\ell}^{\beta_\ell}. \tag{21}$$

The exponents in (21) sum to one. We now analyse their signs.

Assume first that $t > 0$ and all support points are smaller than 1. Since $\omega(\alpha) > 0$ for $0 < \alpha < 1$, we have $\beta_\ell > 0$. Moreover, condition (15) is equivalent to

$$\sum_{\ell=1}^{N} \beta_\ell = t \int \omega(\alpha)\,\mathrm{d}\tau(\alpha) \le 1, \qquad \text{that is,} \qquad \overline{\beta} \ge 0. \tag{22}$$

Hence all exponents of the outer monomial are nonnegative. By Lemma 1.1, every inner function $\|\cdot\|_{\alpha_\ell}$ is concave, while $\|\cdot\|_1$ is linear. By Lemma 1.2, the outer monomial is concave and increasing in all of its arguments. Therefore, for $\lambda \in [0,1]$,

$$\begin{aligned}
&\Phi_{t,\tau}(\lambda\vec{x} + (1-\lambda)\vec{z}) \\
&\quad \ge (\lambda\|\vec{x}\|_1 + (1-\lambda)\|\vec{z}\|_1)^{\overline{\beta}} \prod_{\ell=1}^{N} (\lambda\|\vec{x}\|_{\alpha_\ell} + (1-\lambda)\|\vec{z}\|_{\alpha_\ell})^{\beta_\ell} \\
&\quad \ge \lambda\Phi_{t,\tau}(\vec{x}) + (1-\lambda)\Phi_{t,\tau}(\vec{z}),
\end{aligned} \tag{23}$$

which proves concavity.

Assume next that $t < 0$ and that all support points are smaller than 1. Then every $\beta_\ell < 0$, and consequently

$$\overline{\beta} = 1 - \sum_{\ell=1}^{N} \beta_\ell > 0. \tag{24}$$

Thus $\overline{\beta}$ is the unique positive exponent. The outer monomial is convex, increasing in its first argument, and decreasing in all remaining arguments. Since the power means with $\alpha_\ell < 1$ are concave, we obtain

$$\begin{aligned}
&\Phi_{t,\tau}(\lambda\vec{x} + (1-\lambda)\vec{z}) \\
&\quad \le (\lambda\|\vec{x}\|_1 + (1-\lambda)\|\vec{z}\|_1)^{\overline{\beta}} \prod_{\ell=1}^{N} (\lambda\|\vec{x}\|_{\alpha_\ell} + (1-\lambda)\|\vec{z}\|_{\alpha_\ell})^{\beta_\ell} \\
&\quad \le \lambda\Phi_{t,\tau}(\vec{x}) + (1-\lambda)\Phi_{t,\tau}(\vec{z}).
\end{aligned} \tag{25}$$

No additional integral condition is required in this case.

Finally, assume that $t < 0$ and that $\alpha_* > 1$ is the unique support point above 1. The exponent corresponding to α_* is positive because both t and $\omega(\alpha_*)$ are negative. Every exponent corresponding to a point below 1 is negative. Moreover, (16) implies

$$t \int \omega(\alpha)\,\mathrm{d}\tau(\alpha) \ge 1, \tag{26}$$

where the inequality reverses because $t < 0$. Hence

$$\overline{\beta} = 1 - t \int \omega(\alpha)\,\mathrm{d}\tau(\alpha) \leq 0. \tag{27}$$

The exponent belonging to α_* is therefore the unique positive exponent. Its inner function is convex, whereas every inner function carrying a negative exponent is concave or linear. The monotonicity of the outer monomial in each coordinate then gives the same two-step inequality as in (25), and proves convexity. The argument includes $\alpha_* = +\infty$ by (19) and the convexity of $\|\cdot\|_\infty$.

It remains to discuss atoms at 0 and 1. We treat them by pointwise approximation, which avoids any separate and potentially misleading manipulation of the support factor e^{tH_0}.

An atom at 1 can occur only in item 2(a). Replace it by an atom of the same mass at $1 - 1/n$. Every approximating function is convex by the preceding argument, and it converges pointwise to the desired function because $H_{1-1/n}(p) \to H_1(p)$ for every finite probability vector p. Passing to the limit in the convexity inequality proves the claim.

For an atom at 0, let $a_0 := \tau(\{0\})$ and replace $a_0\delta_0$ by $a_0\delta_{\varepsilon_n}$, where $\varepsilon_n \downarrow 0$. Write

$$I := \int_{[0,+\infty]\setminus\{0,1\}} \omega(\alpha)\,\mathrm{d}\tau(\alpha), \qquad I_n := I + a_0\omega(\varepsilon_n), \qquad \delta_n := I_n - I \downarrow 0. \tag{28}$$

In the cases with a moment constraint, define at the same time

$$t_n := \frac{t}{1 + t\delta_n}. \tag{29}$$

For all sufficiently large n, the denominator is positive and $t_n \to t$. Moreover,

$$t_n I_n - 1 = \frac{tI - 1}{1 + t\delta_n}. \tag{30}$$

Thus $t_n I_n \leq 1$ in the positive case and $t_n I_n \geq 1$ in the negative exceptional case. Every approximating function therefore has the required curvature. In the negative case supported entirely below 1, no moment constraint is present and we simply keep $t_n = t$. Since $H_{\varepsilon_n}(p) \to H_0(p)$ for every finite probability vector p, the approximating functions converge pointwise to $\Phi_{t,\tau}$. Pointwise limits preserve concavity and convexity, so the endpoint $\alpha = 0$ is included. The cases in which one of the vectors is zero follow from positive homogeneity. This completes the proof. □

C. Sufficient conditions for the temperate homomorphisms – general measure

This subsection supplies the full proof of Proposition 1.4 of the concise companion.

We now extend the previous result to an arbitrary probability measure. The main point is to construct a finitely supported approximation on the parameter space which is independent of the vector $\vec{x}$. This permits us to use the same approximating function at $\vec{x}$, $\vec{z}$, and $\lambda\vec{x} + (1-\lambda)\vec{z}$. The required approximation and the associated dominated-convergence argument are recorded in Appendix A 1; we include the complete calculation below because it is central to the passage from the discrete to the general case.

Proposition 1.4. *Let τ be a probability measure on $[0,+\infty]$ and let $t \in \mathbb{R}\setminus\{0\}$. The function in (2) is*

1. *concave if $t > 0$ and*

$$\operatorname{supp}(\tau) \subseteq [0,1], \qquad \tau(\{1\}) = 0, \qquad \int_{[0,1)} \frac{\alpha}{1-\alpha}\,\mathrm{d}\tau(\alpha) \leq \frac{1}{t}; \tag{31}$$

2. convex if $t<0$ and either

(a) $\operatorname{supp}(\tau) \subseteq [0,1]$*; or*

(b) there exists $\alpha_* \in (1,+\infty]$ such that

$$\begin{aligned} &\operatorname{supp}(\tau) \subseteq [0,1] \cup \{\alpha_*\}, && \tau(\{1\}) = 0, \\ &\int_{[0,1)} \frac{\alpha}{1-\alpha}\, \mathrm{d}\tau(\alpha) < +\infty, && \int_{(1,+\infty]} \frac{-\alpha}{1-\alpha}\, \mathrm{d}\tau(\alpha) < +\infty, \\ &\int_{[0,+\infty]\setminus\{1\}} \frac{\alpha}{1-\alpha}\, \mathrm{d}\tau(\alpha) \le \frac{1}{t}. \end{aligned} \tag{32}$$

The condition $\tau(\{1\})=0$ excludes an atom at 1; it does not exclude 1 from the topological support. In the exceptional branch the two nonnegative one-sided integrals are required to be finite before their signed difference is used. A nonatomic measure may still accumulate at 1.

Proof. We first prove item 1. For $n \in \mathbb{N}$, define the finite-valued Borel map

$$T_n(\alpha) := \frac{\lfloor n\alpha \rfloor}{n} \quad (0 \le \alpha < 1), \qquad T_n(1) := 1 - \frac{1}{n}, \tag{33}$$

and let

$$\tau_n := (T_n)_\# \tau \tag{34}$$

be the push-forward measure. Since $\tau(\{1\}) = 0$, the value assigned at 1 does not affect any integral. The function ω is increasing on $[0,1)$, and $T_n(\alpha) \le \alpha$. Therefore

$$\begin{aligned} \int \omega(\beta)\, \mathrm{d}\tau_n(\beta) &= \int \omega(T_n(\alpha))\, \mathrm{d}\tau(\alpha) \\ &\le \int \omega(\alpha)\, \mathrm{d}\tau(\alpha) \le \frac{1}{t}. \end{aligned} \tag{35}$$

Each τ_n is finitely supported and satisfies the hypotheses of Lemma 1.3; hence Φ_{t,τ_n} is concave.

Let p be any probability vector with at most d nonzero entries. The map $\alpha \mapsto H_\alpha(p)$ is continuous on the compactified interval $[0,+\infty]$, and

$$0 \le H_\alpha(p) \le \log d \qquad \text{for all } \alpha \in [0,+\infty]. \tag{36}$$

Since $T_n(\alpha) \to \alpha$ for τ-almost every α, the dominated convergence theorem applies and gives

$$\int H_\beta(p)\, \mathrm{d}\tau_n(\beta) = \int H_{T_n(\alpha)}(p)\, \mathrm{d}\tau(\alpha) \longrightarrow \int H_\alpha(p)\, \mathrm{d}\tau(\alpha). \tag{37}$$

Consequently, $\Phi_{t,\tau_n}(\vec{x}) \to \Phi_{t,\tau}(\vec{x})$ for every $\vec{x}$. Fix $\vec{x}, \vec{z}$ and $\lambda \in [0,1]$. Since the same τ_n occurs at all three vectors, we have

$$\Phi_{t,\tau_n}(\lambda\vec{x} + (1-\lambda)\vec{z}) \ge \lambda \Phi_{t,\tau_n}(\vec{x}) + (1-\lambda)\Phi_{t,\tau_n}(\vec{z}). \tag{38}$$

Taking $n \to \infty$ proves concavity of $\Phi_{t,\tau}$.

For item 2(a), use the same definition on $[0,1)$ but set $T_n(1) := 1$. The measures τ_n are finitely supported in $[0,1]$, so Lemma 1.3, item 2(a), applies without a moment condition. Equation (37) and passage to the pointwise limit prove convexity.

For item 2(b), keep the exceptional point fixed and discretise only the part below 1:

$$T_n(\alpha_*) := \alpha_*, \qquad T_n(\alpha) := \frac{\lfloor n\alpha \rfloor}{n} \quad (0 \le \alpha < 1), \qquad T_n(1) := 1 - \frac{1}{n}. \tag{39}$$

The positive contribution of the part below 1 can only decrease, while the negative contribution of the atom at α_* is unchanged. Thus

$$\int \omega \, \mathrm{d}\tau_n \le \int \omega \, \mathrm{d}\tau \le \frac{1}{t}. \tag{40}$$

Every approximating function is convex by Lemma 1.3, item 2(b). The same dominated-convergence calculation then proves convexity of the limiting function. □

D. Monotonicity under doubly stochastic maps

The previous subsections establish the shape property required for maps acting on the conditioning system. We now verify explicitly the remaining monotonicity under doubly stochastic maps acting on the first system.

Let D be a doubly stochastic matrix and let p be a probability vector. Then p majorises Dp. Every Rényi entropy is Schur concave, including the endpoint values $\alpha = 0, 1, +\infty$, and therefore

$$H_\alpha(Dp) \ge H_\alpha(p) \qquad \text{for every } \alpha \in [0, +\infty]. \tag{41}$$

If $\vec{x}$ is unnormalised, doubly stochasticity also preserves $\|\vec{x}\|_1$. Hence

$$\Phi_{t,\tau}(D\vec{x}) \ge \Phi_{t,\tau}(\vec{x}), \qquad t > 0, \tag{42}$$

$$\Phi_{t,\tau}(D\vec{x}) \le \Phi_{t,\tau}(\vec{x}), \qquad t < 0. \tag{43}$$

The second inequality has precisely the required orientation in the oppositely ordered codomain $\mathbb{R}_+^{\mathrm{op}}$. Combining (42)–(43) with Proposition 1.4 and the lift proved in Section 1 A establishes monotonicity under all conditionally mixing channels in the stated parameter ranges.

E. Sufficient conditions for tropical homomorphisms

This subsection proves Propositions 1.5–1.6 and Remark 1.7 of the concise companion.

We next consider the tropical homomorphisms. The negative tropical quantity is obtained from the temperate one through the limit $t \to -\infty$. The only scalar fact needed is the following finite log-sum-exp limit. If $p_y \ge 0$, $\sum_y p_y = 1$, and $a_y \in \mathbb{R}$, then, with

$$m := \min_{y : p_y > 0} a_y, \tag{44}$$

we have for $t < 0$

$$m \le \frac{1}{t} \log \sum_y p_y e^{t a_y} \le m + \frac{\log p_{y_*}}{t}, \tag{45}$$

where y_* is any minimiser. Both bounds converge to m as $t \to -\infty$.

Proposition 1.5. *Let τ be a probability measure on $[0,+\infty]$. The function*

$$P_{XY} \longmapsto \min_{y:P_Y(y)>0} \int_{[0,+\infty]} H_\alpha(P_{X|Y=y})\,\mathrm{d}\tau(\alpha) \tag{46}$$

is monotone under conditionally mixing channels if either

1. $\mathrm{supp}(\tau) \subseteq [0,1]$*; or*
2. *there exists $\alpha_* \in (1,+\infty]$ such that*

$$\begin{aligned} &\mathrm{supp}(\tau) \subseteq [0,1] \cup \{\alpha_*\}, && \tau(\{1\}) = 0, \\ &\int_{[0,1)} \frac{\alpha}{1-\alpha}\,\mathrm{d}\tau(\alpha) < +\infty, && \int_{(1,+\infty]} \frac{-\alpha}{1-\alpha}\,\mathrm{d}\tau(\alpha) < +\infty, \\ &\int_{[0,+\infty]\setminus\{1\}} \frac{\alpha}{1-\alpha}\,\mathrm{d}\tau(\alpha) \le 0. \end{aligned} \tag{47}$$

Proof. In the first case, Proposition 1.4, item 2(a), gives a monotone temperate homomorphism for every $t < 0$. Apply $(1/t)\log$ to the corresponding monotonicity inequality and then use (45) for the finitely many values of y. Passing to the limit $t \to -\infty$ proves monotonicity of (46).

In the second case, set

$$I(\tau) := \int_{[0,+\infty]\setminus\{1\}} \frac{\alpha}{1-\alpha}\,\mathrm{d}\tau(\alpha). \tag{48}$$

If $I(\tau) < 0$, then $I(\tau) \le 1/t$ for every sufficiently large negative t, and Proposition 1.4, item 2(b), applies. The same limiting argument proves the result.

The boundary case $I(\tau) = 0$ does not follow by fixing τ and applying Proposition 1.4 for finite t, because $1/t < 0$. For $\varepsilon \in (0,1)$, define

$$\tau_\varepsilon := (1-\varepsilon)\tau + \varepsilon\delta_{\alpha_*}. \tag{49}$$

Then

$$I(\tau_\varepsilon) = \varepsilon\frac{\alpha_*}{1-\alpha_*} < 0, \tag{50}$$

where the value is $-\varepsilon$ if $\alpha_* = +\infty$. Hence the tropical quantity corresponding to τ_ε is monotone. For every fixed finite joint distribution, all conditional integrals converge as $\varepsilon \downarrow 0$, and the minimum of finitely many real numbers is continuous. Passing to this limit proves the case $I(\tau) = 0$. □

Proposition 1.6. *The function*

$$P_{XY} \longmapsto \max_{y:P_Y(y)>0} H_0(P_{X|Y=y}) \tag{51}$$

is monotone under conditionally mixing channels.

Proof. Choose $\tau = \delta_0$. The moment in Proposition 1.4, item 1, is zero, so the associated temperate homomorphism is monotone for every $t > 0$. Applying $(1/t)\log$ and using the positive log-sum-exp analogue of (45) gives

$$\lim_{t\to+\infty} \frac{1}{t}\log \Phi_{t,\delta_0}(P_{XY}) = \max_{y:P_Y(y)>0} H_0(P_{X|Y=y}). \tag{52}$$

Taking the limit in the monotonicity inequality proves the claim. □

Remark 1.7. If one fixed τ satisfies all conditions in Proposition 1.4, item 1, for arbitrarily large t, then

$$\int_{[0,1)} \frac{\alpha}{1-\alpha}\,\mathrm{d}\tau(\alpha) = 0. \tag{53}$$

The same conditions include $\tau(\{1\}) = 0$. Since the displayed integrand is strictly positive on $(0,1)$, they force $\tau = \delta_0$.

F. Sufficient conditions for derivations

This subsection proves Proposition 1.8 of the concise companion.

We finally consider the derivations. Although they can be obtained as first-order limits of the temperate homomorphisms, a direct proof is both shorter and more transparent.

Proposition 1.8. *Let τ be a finite positive measure supported on $[0,1]$. Then*

$$\Delta_\tau(P_{XY}) := \sum_{y\in\mathcal{Y}} P_Y(y) \int_{[0,1]} H_\alpha(P_{X|Y=y})\,\mathrm{d}\tau(\alpha) \tag{54}$$

is monotone under conditionally mixing channels. In particular,

$$H_{0,\alpha}(P_{XY}) := \sum_{y\in\mathcal{Y}} P_Y(y) H_\alpha(P_{X|Y=y}) \tag{55}$$

is monotone for every $\alpha \in [0,1]$.

Proof. For $\alpha \in [0,1]$, the Rényi entropy H_α is concave on the probability simplex. At $\alpha = 0$, for $\lambda \in (0,1)$ we have

$$\mathrm{supp}(\lambda p + (1-\lambda)q) = \mathrm{supp}(p) \cup \mathrm{supp}(q). \tag{56}$$

It follows that

$$H_0(\lambda p + (1-\lambda)q) \geq \max\{H_0(p), H_0(q)\} \geq \lambda H_0(p) + (1-\lambda)H_0(q). \tag{57}$$

For $\vec{x} \neq 0$, define the perspective

$$f_\alpha(\vec{x}) := \|\vec{x}\|_1 H_\alpha(\widehat{x}). \tag{58}$$

Let $\vec{x}, \vec{z} \neq 0$, let $\lambda \in [0,1]$, and put

$$s := \lambda\|\vec{x}\|_1 + (1-\lambda)\|\vec{z}\|_1, \qquad \theta := \frac{\lambda\|\vec{x}\|_1}{s}. \tag{59}$$

Then

$$\widehat{\lambda\vec{x} + (1-\lambda)\vec{z}} = \theta\widehat{x} + (1-\theta)\widehat{z}. \tag{60}$$

Using concavity of H_α and multiplying by s gives

$$\begin{aligned} f_\alpha(\lambda\vec{x} + (1-\lambda)\vec{z}) &\geq s\big(\theta H_\alpha(\widehat{x}) + (1-\theta)H_\alpha(\widehat{z})\big) \\ &= \lambda f_\alpha(\vec{x}) + (1-\lambda) f_\alpha(\vec{z}). \end{aligned} \tag{61}$$

The cases involving the zero vector follow from positive homogeneity. Integrating (61) against the positive measure τ preserves the inequality. Section 1 A identifies this concavity property, together with (41), with monotonicity under conditionally mixing channels. Notice that the required property is concavity, not convexity. ∎

The results of this section provide a sufficient parameter region for all temperate, tropical, and derivation cases. In the next section, we construct explicit counterexamples outside this region and prove that the conditions are also necessary without any auxiliary regularity assumption at $\alpha = 1$.

2. NECESSARY CONDITIONS FOR THE PARAMETERS

In this section, we prove that the sufficient conditions obtained in Section 1 are necessary. The proof removes the regularity assumption previously imposed near $\alpha = 1$. In particular, we do not assume that the two one-sided integrals of $|\alpha/(1-\alpha)|$ are finite. Whenever such an integral appears in the final parameter region, its finiteness will be derived from monotonicity.

A. Proof protocol: endpoint differentiation and signed moments

All localizers below use only bounded truncations $[0, a)$ and $(b, +\infty]$, so their coefficients are finite without a singular-moment assumption. In an exceptional upper-support branch, support isolation first makes the upper one-sided moment finite; the truncated curvature inequalities then bound the lower one-sided moment. The signed moment is formed only after these two conclusions. Differentiation under the entropy integral itself needs no singular moment because the removable Shannon quotient is kept intact at fixed dimension.

The difficulty is that the normalised Rényi entropy contains a cancellation at $\alpha = 1$. If its two terms are separated before differentiation, the factor $\alpha/(1-\alpha)$ appears singular and one is led to impose an unnecessary integrability assumption. We therefore proceed in the following order:

$$\boxed{\text{differentiate at fixed finite dimension} \;\longrightarrow\; \text{integrate in } \alpha \;\longrightarrow\; \text{take the high-dimensional limit.}} \tag{62}$$

Every interchange in (62) is justified in Appendix A. The appendix proves joint continuity of the first two derivatives through $\alpha = 1$, gives a complete derivative–integral interchange at fixed dimension, and establishes dimension-independent bounds for the block perturbations used below. The limiting piecewise-affine power envelope is never differentiated.

For notational convenience, we combine the parameter t and the probability measure τ into the finite one-signed measure

$$\mu := t\tau. \tag{63}$$

For a finite signed measure μ, the notation $|\mu|$ means its total-variation measure, and $\operatorname{supp}(\mu) := \operatorname{supp}(|\mu|)$. A point r is called $|\mu|$-null when $|\mu|(\{r\}) = 0$. These conventions are used whenever a localization threshold is chosen below. Thus $\mu \geq 0$ in the $\mathbb{R}_+$ case and $\mu \leq 0$ in the $\mathbb{R}_+^{\mathrm{op}}$ case. We write

$$\Phi_\mu(\vec{x}) := \|\vec{x}\|_1 \exp\left(\int_{[0,+\infty]} H_\alpha(\widehat{x}) \, \mathrm{d}\mu(\alpha) \right). \tag{64}$$

For the negative tropical case, it is convenient to remove the outer norm and exponential and define

$$G_\mu(\vec{x}) := \int_{[0,+\infty]} H_\alpha(\widehat{x}) \, \mathrm{d}\mu(\alpha). \tag{65}$$

Since $0 \leq H_\alpha(\widehat{x}) \leq \log d$ for a d-dimensional vector, both quantities are well defined for every finite measure μ.

Let us also clarify why the perturbation directions used below are admissible. The conditional-majorization semiring contains all finite nonnegative matrices, and in particular every positive one-column vector. If $\vec{x} \in \mathbb{R}^d_{>0}$ and $\vec{v} \in \mathbb{R}^d$, then

$$\vec{x} + \lambda\vec{v} \in \mathbb{R}^d_{>0} \qquad \text{whenever} \qquad |\lambda| < \min_{i:v_i \neq 0} \frac{x_i}{|v_i|}. \tag{66}$$

The direction $\vec{v}$ need not itself be nonnegative and need not be generated by a conditional mixing channel: it is a tangent direction in the cone on which concavity, convexity, or quasi-convexity is tested. The dimension and the direction may depend on an auxiliary integer d. A failure of the required curvature property at any sufficiently large but finite d is already a valid semiring counterexample.

Whenever several coordinate blocks are placed in the same column, we use the notation

$$\mathrm{col}(z^{(1)}, \ldots, z^{(m)}) \tag{67}$$

for their vertical concatenation. This must not be confused with the semiring direct sum $\oplus$, which would create additional conditioning columns and would change the homomorphism into a sum or a tropical maximum over those columns. We also write

$$\mathbf{1}_n := (1, \ldots, 1) \in \mathbb{R}^n \tag{68}$$

for the all-ones vector. When the subscript is a parameter set rather than an integer, $\mathbf{1}_E$ denotes the indicator of E. As usual, $\lceil s \rceil$ is the least integer not smaller than s.

The proof is organised around three elementary high-dimensional perturbations.

1. A two-block perturbation isolates one interval of Rényi parameters and gives the shortest counterexamples for positive measures and for derivations.

2. A three-block perturbation isolates two disjoint intervals. It is needed when convexity can fail because two independent coefficients are positive, and in particular to rule out two separated portions of support above 1.

3. A Shannon-point perturbation produces a bounded first-derivative spike at $\alpha = 1$. It detects an atom at 1 while making the contribution of every nonatomic measure in a neighbourhood of 1 vanish in the high-dimensional limit.

We present the limiting formulas in the main text and use them to give transparent counterexamples. Their complete derivation, including all uniform estimates and applications of dominated convergence, is contained in Appendix A 3 and Appendix A 4.

B. Endpoint-safe localized witness families

a. The two-block perturbation. Fix $r > 0$ with $r \neq 1$, put $R := r + 1$, and let

$$n_{0,d} := \lceil d^R \rceil, \qquad n_{1,d} := \lceil d^{R-r} \rceil. \tag{69}$$

For bounded velocities u_0, u_1, define

$$\vec{x}_r^{(d)}(\lambda) := \mathrm{col}\big((1 + u_0\lambda)\mathbf{1}_{n_{0,d}}, d(1 + u_1\lambda)\mathbf{1}_{n_{1,d}}\big)\,. \tag{70}$$

The power-sum exponents, after removal of the common factor d^R, are

$$0 \qquad \text{and} \qquad \alpha - r. \tag{71}$$

Thus the first block dominates for $\alpha < r$ and the second for $\alpha > r$.

Choose r so that $|\mu|(\{r\}) = 0$. If $r > 1$, define

$$A_r := \int_{(r,+\infty]} \omega(\alpha)\,\mathrm{d}\mu(\alpha). \tag{72}$$

Since the block containing $\alpha = 1$ is the first block, Appendix A 3 b proves

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} \log \Phi_\mu(\vec{x}_r^{(d)}(\lambda))\Big|_{\lambda=0} \longrightarrow (1 - A_r)u_0 + A_r u_1, \tag{73}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}\lambda^2} \log \Phi_\mu(\vec{x}_r^{(d)}(\lambda))\Big|_{\lambda=0} \longrightarrow -(1 - A_r)u_0^2 - A_r u_1^2. \tag{74}$$

If $0 < r < 1$, define instead

$$B_r := \int_{[0,r)} \omega(\alpha)\,\mathrm{d}\mu(\alpha). \tag{75}$$

Now the second block contains $\alpha = 1$, and

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} \log \Phi_\mu(\vec{x}_r^{(d)}(\lambda))\Big|_{\lambda=0} \longrightarrow B_r u_0 + (1 - B_r)u_1, \tag{76}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}\lambda^2} \log \Phi_\mu(\vec{x}_r^{(d)}(\lambda))\Big|_{\lambda=0} \longrightarrow -B_r u_0^2 - (1 - B_r)u_1^2. \tag{77}$$

The convergence in these formulas is locally uniform in the velocities.

b. The three-block perturbation. Fix $0 < a < b < +\infty$ with $a, b \neq 1$, put $R := a + b + 1$, and define

$$\vec{x}_{a,b}^{(d)}(\lambda) := \mathrm{col}\Big((1 + u_0\lambda)\mathbf{1}_{\lceil d^R \rceil}, d(1 + u_1\lambda)\mathbf{1}_{\lceil d^{R-a} \rceil}, d^2(1 + u_2\lambda)\mathbf{1}_{\lceil d^{R-a-b} \rceil}\Big). \tag{78}$$

The three power-sum exponents are

$$q_0(\alpha) = 0, \qquad q_1(\alpha) = \alpha - a, \qquad q_2(\alpha) = 2\alpha - a - b. \tag{79}$$

Consequently, blocks $0, 1, 2$ dominate respectively on

$$[0, a), \qquad (a, b), \qquad (b, +\infty]. \tag{80}$$

Let k be the index of the cell containing 1, and assume $|\mu|(\{a, b\}) = 0$. For $j \neq k$, put

$$A_j := \int_{J_j} \omega(\alpha)\,\mathrm{d}\mu(\alpha), \tag{81}$$

where J_j denotes the corresponding interval in (80). No integral is introduced over the cell containing 1. Then

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} \log \Phi_\mu(\vec{x}_{a,b}^{(d)}(\lambda))\Big|_{\lambda=0} \longrightarrow \left(1 - \sum_{j\neq k} A_j\right) u_k + \sum_{j\neq k} A_j u_j, \tag{82}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}\lambda^2} \log \Phi_\mu(\vec{x}_{a,b}^{(d)}(\lambda))\Big|_{\lambda=0} \longrightarrow -\left(1 - \sum_{j\neq k} A_j\right) u_k^2 - \sum_{j\neq k} A_j u_j^2. \tag{83}$$

Again, the convergence is locally uniform in (u_0, u_1, u_2).

c. The Shannon-point perturbation. Fix a $|\mu|$-null point $c > 1$, put $p = q := 1/2$ and $R := c + 2$, and define

$$n_{0,d} := \lceil d^R \rceil, \qquad n_{1,d} := \lceil d^{R-1} \rceil, \qquad n_{2,d} := \lceil d^{R-1-c} \rceil. \tag{84}$$

The finite path is

$$\vec{x}^{(d)}_{\mathrm{Sh}}(\lambda) := \mathrm{col}\left(\left(q + \frac{h\lambda}{\log d} \right) \mathbf{1}_{n_{0,d}}, d \left(p - \frac{h\lambda}{\log d} \right) \mathbf{1}_{n_{1,d}}, d^2 (1 + v\lambda) \mathbf{1}_{n_{2,d}} \right). \tag{85}$$

This is literally the rounded three-block Shannon family used in Lean, with multiplicative block velocities $h/(q \log d)$, $-h/(p \log d)$, and v. The power-sum exponents are

$$c + 2, \qquad c + 1 + \alpha, \qquad 1 + 2\alpha. \tag{86}$$

The first block dominates for $\alpha < 1$, the second for $1 < \alpha < c$, and the third for $\alpha > c$. The first two blocks tie at $\alpha = 1$; their $1/\log d$ perturbation is chosen precisely so that the Shannon derivative remains of order one.

Set

$$m := \mu(\{1\}), \qquad A_c := \int_{(c,+\infty]} \omega(\alpha) \, \mathrm{d}\mu(\alpha). \tag{87}$$

Appendix A 4 proves

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} \log \Phi_\mu(\vec{x}^{(d)}_{\mathrm{Sh}}(\lambda)) \bigg|_{\lambda=0} \longrightarrow mh + A_c v, \tag{88}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}\lambda^2} \log \Phi_\mu(\vec{x}^{(d)}_{\mathrm{Sh}}(\lambda)) \bigg|_{\lambda=0} \longrightarrow -A_c v^2. \tag{89}$$

The first derivative produced by the nonatomic part of the measure in a full neighbourhood of 1 tends to zero; only the atom m survives. This is the step that removes the need for an integrability assumption at $\alpha = 1$.

We now apply these three perturbations. Recall that, along any affine line, if

$$\Lambda_d(\lambda) := \log \Phi_\mu(\vec{x}^{(d)}(\lambda)), \tag{90}$$

then

$$\frac{\Phi''_\mu(0)}{\Phi_\mu(0)} = \Lambda''_d(0) + \big(\Lambda'_d(0)\big)^2. \tag{91}$$

Thus a positive value in (91) contradicts concavity, while a negative value contradicts convexity whenever the first derivative is zero.

C. Necessary conditions; temperate case

Proposition 2.1. *Let μ be an arbitrary finite positive measure. The function in (64) can be concave only if*

$$\mu(\{1\}) = 0, \qquad \mathrm{supp}(\mu) \subseteq [0,1], \qquad \int_{[0,1)} \frac{\alpha}{1-\alpha} \, \mathrm{d}\mu(\alpha) \le 1. \tag{92}$$

The integral is an extended nonnegative integral; the conclusion shows that it is finite.

Proof. We prove the three statements in (92) separately.

First, suppose $m := \mu(\{1\}) > 0$. In the Shannon perturbation choose

$$h = 1, \qquad v = 0. \tag{93}$$

Equations (88)–(89) give

$$\Lambda_d'(0) \longrightarrow m, \qquad \Lambda_d''(0) \longrightarrow 0. \tag{94}$$

Therefore

$$\frac{\Phi_\mu''(0)}{\Phi_\mu(0)} \longrightarrow m^2 > 0. \tag{95}$$

For every sufficiently large but finite d, the line (85) violates concavity. Hence $\mu(\{1\}) = 0$.

Second, suppose that $\operatorname{supp}(\mu)$ intersects $(1, +\infty]$. Choose a μ-null number $r > 1$ such that $\mu((r, +\infty]) > 0$. Then

$$A_r = \int_{(r,+\infty]} \omega(\alpha)\, \mathrm{d}\mu(\alpha) < 0. \tag{96}$$

Use the two-block perturbation with

$$u_0 = 0, \qquad u_1 = 1. \tag{97}$$

Equations (73)–(74) yield

$$\Lambda_d'(0) \longrightarrow A_r, \qquad \Lambda_d''(0) \longrightarrow -A_r. \tag{98}$$

Consequently,

$$\frac{\Phi_\mu''(0)}{\Phi_\mu(0)} \longrightarrow A_r^2 - A_r = A_r(A_r - 1) > 0, \tag{99}$$

again contradicting concavity. Hence the measure is supported on $[0, 1]$.

Finally, choose a μ-null number $r \in (0, 1)$ and define B_r as in (75). Use

$$u_0 = 1, \qquad u_1 = 0. \tag{100}$$

Then (76)–(77) give

$$\Lambda_d'(0) \longrightarrow B_r, \qquad \Lambda_d''(0) \longrightarrow -B_r, \tag{101}$$

so concavity requires

$$B_r(B_r - 1) \leq 0. \tag{102}$$

Since $B_r \geq 0$, this gives $B_r \leq 1$. Let $r \uparrow 1$ through μ-null points. The functions $\omega \mathbf{1}_{[0,r)}$ increase pointwise to $\omega \mathbf{1}_{[0,1)}$, and the monotone convergence theorem yields

$$\int_{[0,1)} \omega(\alpha)\, \mathrm{d}\mu(\alpha) = \lim_{r \uparrow 1} B_r \leq 1. \tag{103}$$

This proves the proposition. □

We now consider a finite negative measure μ. The first obstruction is an atom at the Shannon point together with any support above 1.

Proposition 2.2. *Let μ be an arbitrary finite negative measure such that $\mu(\{1\}) \neq 0$. If the function in (64) is convex, then*

$$\operatorname{supp}(\mu) \subseteq [0, 1]. \tag{104}$$

Proof. Suppose that the measure has support above 1. Choose a $|\mu|$-null point $c > 1$ with $\mu((c, +\infty]) < 0$. Since $\omega < 0$ above 1,

$$A_c := \int_{(c,+\infty]} \omega(\alpha)\, \mathrm{d}\mu(\alpha) > 0. \tag{105}$$

Write $m := \mu(\{1\}) < 0$ and choose in (85)

$$v = 1, \qquad h = -\frac{A_c}{m}. \tag{106}$$

Then

$$\Lambda_d'(0) \longrightarrow mh + A_c = 0, \qquad \Lambda_d''(0) \longrightarrow -A_c < 0. \tag{107}$$

By local uniformity of the convergence, the second derivative of Φ_μ is negative for every sufficiently large finite d, contradicting convexity. This proves (104). $\square$

If there is no atom at 1, we cannot initially define a global coefficient $1 - \int \omega\, \mathrm{d}\mu$, because one or both one-sided integrals may be infinite. We instead work with truncated coefficients. This is the only modification needed to remove the former regularity assumption.

Proposition 2.3. *Let μ be an arbitrary finite negative measure such that $\mu(\{1\}) = 0$, and assume that the function in (64) is convex. Let $a < 1 < b$ be $|\mu|$-null points such that $\mu((b, +\infty]) < 0$, and define*

$$B_a := \int_{[0,a)} \omega(\alpha)\, \mathrm{d}\mu(\alpha) \leq 0, \qquad A_b := \int_{(b,+\infty]} \omega(\alpha)\, \mathrm{d}\mu(\alpha) > 0. \tag{108}$$

Then

$$A_b + B_a \geq 1. \tag{109}$$

In particular, if both one-sided moments exist as finite numbers and

$$\int_{[0,+\infty]\setminus\{1\}} \omega(\alpha)\, \mathrm{d}\mu(\alpha) < 1, \tag{110}$$

then μ cannot have support above 1.

Proof. Use the three-block perturbation with thresholds $a < 1 < b$. The middle block contains $\alpha = 1$. The three coefficients in (82) are

$$B_a, \qquad C := 1 - A_b - B_a, \qquad A_b. \tag{111}$$

Suppose that (109) fails. Then $C > 0$, while $A_b > 0$. Thus the limiting curvature has two positive coefficients. Choose the explicit velocities

$$u_0 = 0, \qquad u_1 = \frac{1}{C}, \qquad u_2 = -\frac{1}{A_b}. \tag{112}$$

The limiting first derivative is

$$B_a u_0 + C u_1 + A_b u_2 = 0 + 1 - 1 = 0, \tag{113}$$

whereas the limiting second derivative is

$$-B_a u_0^2 - C u_1^2 - A_b u_2^2 = -\frac{1}{C} - \frac{1}{A_b} < 0. \tag{114}$$

Therefore $\Phi_\mu''(0) < 0$ for every sufficiently large finite d, contradicting convexity. Hence $C \leq 0$, which is precisely (109).

If the two one-sided moments are finite and their sum is smaller than 1, choose $a \uparrow 1$ and $b \downarrow 1$ through $|\mu|$-null points. The truncated coefficients converge to the corresponding one-sided integrals, so for sufficiently close a and b one has $A_b + B_a < 1$, contradicting the first part. □

Remark 2.4. The point and directions in the preceding propositions are not required to be normalised or traceless. This is not an enlargement of the semiring problem: the semiring is defined on unnormalised nonnegative matrices, and Section 1 A tests the curvature property on the whole cone. Positive homogeneity also shows that rescaling a counterexample does not change the sign of the relevant curvature. Thus the one-column perturbations above are directly admissible.

Remark 2.5. If one nevertheless wishes to realise the same calculation on a fixed-mass affine hyperplane, one may append a second positive column and vary its total mass by the negative of the mass variation of the first column. Additivity of the homomorphism separates the two columns, and the second column contributes only a linear correction. This optional construction leads to the same second-order obstruction, but it is not needed for the semiring argument.

It remains to show that a convex negative-measure homomorphism cannot have two different support points above 1. This is where the three-block perturbation is essential.

Proposition 2.6. *Let μ be an arbitrary finite negative measure. If the function in (64) is convex, then*

$$\operatorname{supp}(\mu) \cap (1, +\infty] \tag{115}$$

contains at most one point.

Proof. Suppose that two distinct points

$$1 < \alpha_1 < \alpha_2 \leq +\infty \tag{116}$$

lie in the support. Choose $|\mu|$-null thresholds

$$1 < a < \alpha_1 < b < \alpha_2. \tag{117}$$

If $\alpha_2 = +\infty$, the last inequality means that b is any finite null point larger than α_1. The first block in (78) contains $\alpha = 1$. Define

$$A_1 := \int_{(a,b)} \omega(\alpha)\, \mathrm{d}\mu(\alpha) > 0, \qquad A_2 := \int_{(b,+\infty]} \omega(\alpha)\, \mathrm{d}\mu(\alpha) > 0. \tag{118}$$

The strict inequalities follow because μ and ω are both negative above 1, while each interval has positive $|\mu|$-measure by the definition of support.

Choose

$$u_0 = 0, \qquad u_1 = \frac{1}{A_1}, \qquad u_2 = -\frac{1}{A_2}. \tag{119}$$

Then the limiting first derivative in (82) is

$$A_1 u_1 + A_2 u_2 = 0, \tag{120}$$

whereas the limiting second derivative is

$$-A_1 u_1^2 - A_2 u_2^2 = -\frac{1}{A_1} - \frac{1}{A_2} < 0. \tag{121}$$

This gives a finite-dimensional violation of convexity for every sufficiently large d. Hence there can be at most one support point above 1. □

We now combine the preceding results. If a convex negative-measure homomorphism has no support above 1, no moment condition is required and an atom at 1 is allowed. Suppose instead that there is support above 1. Proposition 2.6 shows that the restriction of μ to $(1, +\infty]$ is a single atom at some $\alpha_* \in (1, +\infty]$. Proposition 2.2 gives $\mu(\{1\}) = 0$.

Choose $b \in (1, \alpha_*)$ with $|\mu|(\{b\}) = 0$. Then $A_b = \omega(\alpha_*)\mu(\{\alpha_*\})$ is a finite positive number, so the upper one-sided moment is finite. Proposition 2.3 gives

$$B_a \geq 1 - A_b \qquad \text{for every null } a < 1. \tag{122}$$

As $a \uparrow 1$, the quantities B_a form a decreasing family because the integrand $\omega \, \mathrm{d}\mu$ is negative below 1. The lower bound in (122) prevents convergence to $-\infty$. Hence the lower one-sided moment is finite. Only now, after both one-sided finiteness statements have been proved, do we form their signed difference. Taking the limit gives

$$\int_{[0,+\infty]\setminus\{1\}} \omega(\alpha) \, \mathrm{d}\mu(\alpha) \geq 1. \tag{123}$$

Thus finiteness of the weighted integral is a consequence of convexity in the exceptional branch, rather than an assumption.

Returning to $\mu = t\tau$, we obtain the exact temperate parameter conditions

$$t > 0: \quad \operatorname{supp}(\tau) \subseteq [0,1], \qquad \tau(\{1\}) = 0, \qquad \int_{[0,1)} \frac{\alpha}{1-\alpha} \, \mathrm{d}\tau(\alpha) \leq \frac{1}{t}; \tag{124}$$

$$t < 0: \quad \operatorname{supp}(\tau) \subseteq [0,1], \quad \text{or} \quad \begin{cases} \operatorname{supp}(\tau) \subseteq [0,1] \cup \{\alpha_*\}, \\ \tau(\{1\}) = 0, \\ \displaystyle\int_{[0,1)} \frac{\alpha}{1-\alpha} \, \mathrm{d}\tau(\alpha) < +\infty, \\ \displaystyle\int_{(1,+\infty]} \frac{-\alpha}{1-\alpha} \, \mathrm{d}\tau(\alpha) < +\infty, \\ \displaystyle\int_{[0,+\infty]\setminus\{1\}} \frac{\alpha}{1-\alpha} \, \mathrm{d}\tau(\alpha) \leq \frac{1}{t}, \end{cases} \qquad \alpha_* \in (1, +\infty]. \tag{125}$$

In (125), division of (123) by the negative number t reverses the inequality.

D. Necessary conditions for tropical homomorphisms

We now prove Propositions 2.7–2.8 of the concise companion. The negative branch uses the same Shannon and three-block witnesses as the temperate case, with an additional exact stationarity correction. The positive branch uses a separate binary fixed-face witness.

Proposition 2.7. *Let τ be a probability measure on $[0,+\infty]$. If the function in (46) is monotone under conditionally mixing channels, then either*

1. $\operatorname{supp}(\tau) \subseteq [0,1]$*; or*
2. *there is a unique $\alpha_* \in (1,+\infty]$ such that*

$$\begin{aligned} &\operatorname{supp}(\tau) \subseteq [0,1]\cup\{\alpha_*\}, && \tau(\{1\})=0, \\ &\int_{[0,1)} \frac{\alpha}{1-\alpha}\,\mathrm{d}\tau(\alpha) < +\infty, && \int_{(1,+\infty]} \frac{-\alpha}{1-\alpha}\,\mathrm{d}\tau(\alpha) < +\infty, \\ &\int_{[0,+\infty]\setminus\{1\}} \frac{\alpha}{1-\alpha}\,\mathrm{d}\tau(\alpha) \le 0. \end{aligned} \tag{126}$$

Proof. The negative tropical homomorphism is governed by quasi-convexity, not convexity. Apply the one-column shape reduction with the finite negative measure $\mu := -\tau$; multiplying it by another positive scalar would not change the argument. We use only the implication

$$g \text{ is quasi-convex and } C^2, \qquad g'(0)=0 \quad\Longrightarrow\quad g''(0)\ge 0. \tag{127}$$

Stationarity must hold exactly at the finite dimension where quasi-convexity is applied. Appendix A 5 proves (127) and corrects a limiting stationary velocity by a dimension-dependent multiple of a transverse velocity.

For G_μ in (65), the norm coefficient is absent. In the three-block notation the limiting derivatives are

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} G_\mu(\vec{x}^{(d)}_{a,b}(\lambda))\Big|_{\lambda=0} \longrightarrow -\left(\sum_{j\neq k} A_j\right) u_k + \sum_{j\neq k} A_j u_j, \tag{128}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}\lambda^2} G_\mu(\vec{x}^{(d)}_{a,b}(\lambda))\Big|_{\lambda=0} \longrightarrow \left(\sum_{j\neq k} A_j\right) u_k^2 - \sum_{j\neq k} A_j u_j^2. \tag{129}$$

The Shannon path has the limits (88)–(89), since its norm derivatives vanish.

There are three counterexamples, exactly as in Lean. If the signed Shannon atom $m<0$ and upper coefficient $A_c>0$ are both present, use $(h,v)=(-A_c/m,1)$. The limiting derivatives are 0 and $-A_c<0$; the transverse direction $e=(0,1)$ has limiting first derivative $A_c\neq 0$. If two upper portions have coefficients $A_1, A_2>0$, use

$$u=\left(0,\frac{1}{A_1},-\frac{1}{A_2}\right), \qquad e=(0,1,0). \tag{130}$$

The limiting first derivative is zero, the second is $-1/A_1-1/A_2<0$, and the first derivative in direction e is A_1. Thus upper support contains at most one point, and an upper point forces $\tau(\{1\})=0$.

Finally suppose that the unique upper point α_* exists. For $a < 1 < b < \alpha_*$, use $B_a \le 0$ and $A_b > 0$ from (108). The tropical coefficients are

$$B_a, \qquad C := -A_b - B_a, \qquad A_b. \tag{131}$$

If $A_b + B_a < 0$, then $C > 0$, and the exact Lean directions are

$$u = \left(0, \frac{1}{C}, -\frac{1}{A_b}\right), \qquad e = (0, 0, 1). \tag{132}$$

The limiting first derivative along u is zero, the second derivative is

$$-\frac{1}{C} - \frac{1}{A_b} < 0, \tag{133}$$

and the first derivative along e is $A_b \ne 0$. The correction proved in Appendix A 5 therefore supplies an exact finite stationary line while preserving the negative second derivative, contradicting quasi-convexity. Hence $A_b + B_a \ge 0$.

The isolated upper atom makes the upper one-sided moment finite. Letting $a \uparrow 1$ proves lower-moment finiteness. Only then do we form the signed moment; after writing $\mu = -k\tau$ with $k > 0$, it satisfies

$$\int_{[0,+\infty]\setminus\{1\}} \frac{\alpha}{1-\alpha} \, \mathrm{d}\tau(\alpha) \le 0. \tag{134}$$

This proves the proposition. Notice that the second derivative was used only after exact finite-dimensional stationarity was imposed. $\square$

Proposition 2.8. *Let τ be a probability measure on $[0, +\infty]$. If*

$$P_{XY} \longmapsto \max_{y : P_Y(y) > 0} \int_{[0,+\infty]} H_\alpha(P_{X|Y=y}) \, \mathrm{d}\tau(\alpha) \tag{135}$$

is monotone under conditionally mixing channels, then $\tau = \delta_0$.

Proof. Put

$$A_\tau(p) := \int_{[0,+\infty]} H_\alpha(p) \, \mathrm{d}\tau(\alpha). \tag{136}$$

Positive-tropical monotonicity implies

$$A_\tau(\lambda p + (1-\lambda)q) \ge \max\{A_\tau(p), A_\tau(q)\}, \qquad 0 < \lambda < 1. \tag{137}$$

If p and q have the same support, each can be expressed as a sufficiently small mixture of the other with a third point on the same face. Applying (137) in both directions shows that A_τ is constant on the relative interior of every simplex face.

Now use the exact binary pair formalized in Lean,

$$p = (1/3, 2/3), \qquad u = (1/2, 1/2). \tag{138}$$

Constancy gives $A_\tau(p) = A_\tau(u) = \log 2$. Yet $H_0(p) = \log 2$ and $H_\alpha(p) < \log 2$ for every $\alpha \in (0, +\infty]$. Therefore

$$0 = \int_{[0,+\infty]} \big(\log 2 - H_\alpha(p)\big) \, \mathrm{d}\tau(\alpha). \tag{139}$$

The integrand is nonnegative and strictly positive on $(0, +\infty]$, so $\tau((0, +\infty]) = 0$ and $\tau = \delta_0$. Together with Proposition 1.6, this proves the equivalence. $\square$

E. Necessary conditions for derivations

We finally prove Propositions 2.9–2.10 of the concise companion. It is useful to treat the full barycentre at once. Let ν be a finite positive measure and define on an unnormalised column

$$D_\nu(\vec{x}) := \|\vec{x}\|_1 \int_{[0,+\infty]} H_\alpha(\widehat{x})\,\mathrm{d}\nu(\alpha). \tag{140}$$

By Section 1 A, monotonicity of the derivation requires concavity of D_ν.

Proposition 2.9. *Let ν be an arbitrary finite positive measure. If the function in (140) is concave, then*

$$\operatorname{supp}(\nu) \subseteq [0,1]. \tag{141}$$

In particular, the extremal function

$$\vec{x} \longmapsto \|\vec{x}\|_1 H_\alpha(\widehat{x}) \tag{142}$$

is concave only if $\alpha \in [0,1]$.

Proof. Suppose that the support of ν intersects $(1,+\infty]$. Choose a ν-null point $r > 1$ such that $\nu((r,+\infty]) > 0$, and set

$$A_r := \int_{(r,+\infty]} \omega(\alpha)\,\mathrm{d}\nu(\alpha) < 0. \tag{143}$$

Use the two-block perturbation (70) with $u_0 = 0$ and $u_1 = 1$. Write

$$S_d(\lambda) := \|\vec{x}_r^{(d)}(\lambda)\|_1, \qquad K_d(\lambda) := \int H_\alpha(\widehat{x_r^{(d)}(\lambda)})\,\mathrm{d}\nu(\alpha). \tag{144}$$

Since S_d is affine,

$$\frac{D_\nu''(0)}{S_d(0)} = 2\frac{S_d'(0)}{S_d(0)}K_d'(0) + K_d''(0). \tag{145}$$

The first block contains $\alpha = 1$ and has velocity zero, so

$$\frac{S_d'(0)}{S_d(0)} \longrightarrow 0. \tag{146}$$

The two-block localisation gives

$$K_d'(0) \longrightarrow A_r, \qquad K_d''(0) \longrightarrow -A_r > 0. \tag{147}$$

Substituting these limits into (145) yields

$$\frac{D_\nu''(0)}{S_d(0)} \longrightarrow -A_r > 0. \tag{148}$$

Thus concavity fails for every sufficiently large finite d. This proves (141). Taking $\nu = \delta_\alpha$ gives the extremal statement. $\square$

Proposition 2.10. *Among the derivations represented in Section 3 of Ref. [1] as positive barycentres of the quantities $H_{0,\alpha}$, the monotone ones are precisely those whose representing measure is supported on $[0,1]$. They have the form*

$$\Delta(P_{XY}) = \int_{[0,1]} H_{0,\alpha}(P_{XY})\, \mathrm{d}\tau(\alpha), \tag{149}$$

where τ is a finite positive measure. Under the normalization adopted in Section 3 of Ref. [1], τ is a probability measure. Hence, within that representation theorem, the cone of monotone derivations is generated by $H_{0,\alpha}$ with $\alpha \in [0,1]$.

Proof. The representation theorem of Section 3 in Ref. [1] is the imported starting point: it represents every derivation at $\|\cdot\|$ as a positive barycentre of the quantities $H_{0,\alpha}$ with $\alpha \in [0,+\infty]$. Proposition 2.9 shows that monotonicity forces the representing measure to be supported on $[0,1]$. Conversely, Proposition 1.8 proves monotonicity for every positive measure supported on $[0,1]$. This establishes (149). The representation theorem itself is not reproved in this companion. □

The counterexamples above are deliberately chosen to be as simple as possible. A two-block vector is sufficient whenever only one forbidden coefficient must be isolated; three blocks are needed only when two independent positive coefficients must be exposed; and the separate Shannon perturbation is used only to detect an atom at $\alpha = 1$. In every case, the limiting curvature is strict. Appendix A 6 explains how Taylor's theorem converts the strict limiting sign into an explicit pair of finite-dimensional semiring elements violating concavity, convexity, or quasi-convexity.

Appendix A: Technical foundations for the concise classification

This appendix collects the analytic statements used in Sections 1 and 2. The purpose of separating them from the main text is twofold. First, it keeps the counterexamples in Section 2 visible: the main argument is a finite-dimensional curvature test with explicitly chosen velocities. Second, it makes precise every operation which involves a parameter integral, a derivative, or a high-dimensional limit. In particular, no appeal is made to an informal interchange of operations.

The technical dependency order is: (1) real-power Hessians and common discretization; (2) removable endpoint differentiation at 0, 1, and $+\infty$; (3) differentiation under a finite signed measure by total-variation domination; (4) compact-uniform two- and three-block localization; (5) the separate Shannon localizer; (6) exact tropical stationarity; and (7) Taylor conversion to finite witnesses followed by the channel lift of Section 1 A. This order explains how the concise proof ideas become formal arguments: the main text chooses the witness and reads its sign, while this appendix verifies that every derivative, integral, limit, and finite correction used to obtain that sign is legitimate.

We fix the technical notation before it is used. If $\rho = (\rho_i)_{i=1}^N$ is a finite probability vector and $f = (f_i)$ and $g = (g_i)$ are real arrays, then

$$\mathbb{E}_\rho[f] := \sum_{i=1}^{N} \rho_i f_i, \tag{A1}$$

$$\mathrm{Var}_\rho(f) := \mathbb{E}_\rho[f^2] - \big(\mathbb{E}_\rho[f]\big)^2, \tag{A2}$$

$$\mathrm{Cov}_\rho(f,g) := \mathbb{E}_\rho[fg] - \mathbb{E}_\rho[f]\,\mathbb{E}_\rho[g]. \tag{A3}$$

An omitted subscript means that the probability weights have been named in the surrounding sentence. We use

$$\mathrm{D}^2 F(\vec{x})[\vec{v},\vec{v}] := \frac{\mathrm{d}^2}{\mathrm{d}s^2} F(\vec{x} + s\vec{v})\bigg|_{s=0} \tag{A4}$$

for the second directional derivative, ∂_i for differentiation in the ith coordinate, δ_{ij} for the Kronecker delta, and $\|f\|_\infty := \max_i |f_i|$ for a finite array. The notation $a_s = O(b_s)$ means that $|a_s| \le C|b_s|$ near the indicated limit, whereas $a_s = o(b_s)$ means $a_s/b_s \to 0$. If auxiliary parameters are present, "locally uniform" means uniform on every compact set of those parameters. Finally, $g \in L^1(\nu)$ means $\int |g|\,\mathrm{d}\nu < +\infty$.

1. Real-power curvature, endpoint continuity, and common discretization

We begin with the two curvature computations used in Lemmas 1.1 and 1.2.

Lemma A.1 (Hessian of the power mean). *Let $\alpha \in (0,+\infty)$ and define, on $\mathbb{R}^d_{>0}$,*

$$F_\alpha(\vec{x}) := \left(\sum_{i=1}^{d} x_i^\alpha\right)^{1/\alpha}. \tag{A5}$$

For every $\vec{v} \in \mathbb{R}^d$,

$$\mathrm{D}^2 F_\alpha(\vec{x})[\vec{v},\vec{v}] = (\alpha-1)\left(\sum_i x_i^\alpha\right)^{1/\alpha-2} \times \left[\left(\sum_i x_i^\alpha\right)\left(\sum_i x_i^{\alpha-2} v_i^2\right) - \left(\sum_i x_i^{\alpha-1} v_i\right)^2\right]. \tag{A6}$$

The expression in square brackets is nonnegative. Consequently, F_α is concave for $0 < \alpha < 1$, linear for $\alpha = 1$, and convex for $\alpha > 1$. These conclusions extend continuously from the strictly positive cone to $\mathbb{R}^d_{\geq 0}$. Moreover, $F_\infty(\vec{x}) := \max_i x_i$ is convex.

Proof. Put

$$S_\alpha(\vec{x}) := \sum_i x_i^\alpha, \qquad F_\alpha = S_\alpha^{1/\alpha}. \tag{A7}$$

Along the affine line $\vec{x} + s\vec{v}$, direct differentiation gives

$$\frac{\mathrm{d}}{\mathrm{d}s} S_\alpha(\vec{x} + s\vec{v})\bigg|_{s=0} = \alpha \sum_i x_i^{\alpha-1} v_i, \tag{A8}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}s^2} S_\alpha(\vec{x} + s\vec{v})\bigg|_{s=0} = \alpha(\alpha-1) \sum_i x_i^{\alpha-2} v_i^2. \tag{A9}$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}s} S_\alpha^{1/\alpha} = \frac{1}{\alpha} S_\alpha^{1/\alpha - 1} S_\alpha', \tag{A10}$$

we obtain

$$\frac{\mathrm{d}^2}{\mathrm{d}s^2} S_\alpha^{1/\alpha} = \frac{1}{\alpha}\left(\frac{1}{\alpha} - 1\right) S_\alpha^{1/\alpha-2} (S_\alpha')^2 + \frac{1}{\alpha} S_\alpha^{1/\alpha-1} S_\alpha''. \tag{A11}$$

Substitution of (A8)–(A9) into (A11), followed by extraction of the common factor $(\alpha-1)S_\alpha^{1/\alpha-2}$, gives exactly (A6).

To determine the sign of the square bracket, apply the Cauchy–Schwarz inequality to the vectors

$$a_i := x_i^{\alpha/2}, \qquad b_i := x_i^{\alpha/2-1} v_i. \tag{A12}$$

It gives

$$\left(\sum_i x_i^{\alpha-1} v_i\right)^2 = \left(\sum_i a_i b_i\right)^2 \leq \left(\sum_i a_i^2\right)\left(\sum_i b_i^2\right) = \left(\sum_i x_i^\alpha\right)\left(\sum_i x_i^{\alpha-2} v_i^2\right). \tag{A13}$$

Thus the bracket is nonnegative, and the sign of the Hessian is the sign of $\alpha - 1$. The extension to zero coordinates follows by replacing $\vec{x}$ by $\vec{x} + \varepsilon\mathbf{1}$, applying the inequality in the strictly positive cone, and letting $\varepsilon \downarrow 0$. Finally, F_∞ is the pointwise maximum of the linear coordinate functions $\vec{x} \mapsto x_i$, and is therefore convex. □

Lemma A.2 (Hessian and sign classification of a monomial)**.** *Let $\beta_0, \dots, \beta_N \in \mathbb{R}$ satisfy $\sum_{j=0}^N \beta_j = 1$, and put*

$$M_\beta(\vec{y}) := \prod_{j=0}^N y_j^{\beta_j}, \qquad \vec{y} \in \mathbb{R}^{N+1}_{>0}. \tag{A14}$$

For every $\vec{v} \in \mathbb{R}^{N+1}$,

$$\frac{\mathrm{D}^2 M_\beta(\vec{y})[\vec{v}, \vec{v}]}{M_\beta(\vec{y})} = \left(\sum_{j=0}^N \beta_j \frac{v_j}{y_j}\right)^2 - \sum_{j=0}^N \beta_j \left(\frac{v_j}{y_j}\right)^2. \tag{A15}$$

It follows that M_β is concave if and only if every β_j is nonnegative, and convex if and only if exactly one coefficient is positive and all the remaining coefficients are nonpositive.

Proof. The logarithm of the monomial is

$$\log M_\beta(\vec{y}) = \sum_j \beta_j \log y_j. \tag{A16}$$

Consequently,

$$\partial_i M_\beta = M_\beta \frac{\beta_i}{y_i}, \tag{A17}$$

and

$$\partial_i \partial_j M_\beta = M_\beta \left(\frac{\beta_i \beta_j}{y_i y_j} - \delta_{ij} \frac{\beta_i}{y_i^2} \right). \tag{A18}$$

Contracting (A18) with $v_i v_j$ proves (A15).

Assume first that all $\beta_j \geq 0$. Since they sum to one, the right-hand side of (A15) is the negative of the variance of the numbers v_j/y_j under the probability weights β_j:

$$\left(\sum_j \beta_j a_j \right)^2 - \sum_j \beta_j a_j^2 = -\operatorname{Var}_\beta(a) \leq 0. \tag{A19}$$

Hence the monomial is concave. Conversely, if the Hessian is negative semidefinite, choose $a_j := v_j/y_j$ with $a_k = 1$ and all other coordinates zero. Then

$$\beta_k^2 - \beta_k = \beta_k(\beta_k - 1) \leq 0, \tag{A20}$$

so $0 \leq \beta_k \leq 1$ for every k. This proves the concavity classification.

Suppose next that exactly one coefficient is positive. Relabel the coordinates so that

$$\beta_0 = 1 + \Gamma > 0, \qquad \beta_j = -\gamma_j \leq 0 \quad (j \geq 1), \qquad \Gamma := \sum_{j \geq 1} \gamma_j. \tag{A21}$$

Writing $a_j = a_0 + d_j$ for $j \geq 1$, a direct calculation gives

$$\begin{aligned} \left(\sum_j \beta_j a_j \right)^2 - \sum_j \beta_j a_j^2 &= \left(a_0 - \sum_{j \geq 1} \gamma_j d_j \right)^2 - \left(a_0^2 - 2a_0 \sum_{j \geq 1} \gamma_j d_j - \sum_{j \geq 1} \gamma_j d_j^2 \right) \\ &= \left(\sum_{j \geq 1} \gamma_j d_j \right)^2 + \sum_{j \geq 1} \gamma_j d_j^2 \geq 0. \end{aligned} \tag{A22}$$

Thus the monomial is convex.

Conversely, positive semidefiniteness and the one-coordinate test imply that each β_j belongs to $(-\infty, 0] \cup [1, +\infty)$. If two coefficients β_i and β_j were positive, choose

$$a_i = \frac{1}{\beta_i}, \qquad a_j = -\frac{1}{\beta_j}, \qquad a_k = 0 \quad (k \neq i, j). \tag{A23}$$

Then $\sum_k \beta_k a_k = 0$, whereas

$$\sum_k \beta_k a_k^2 = \frac{1}{\beta_i} + \frac{1}{\beta_j} > 0, \tag{A24}$$

so (A15) is negative. This contradicts convexity. Since the coefficients sum to one, at least one is positive, and the convexity classification follows. □

We next record the elementary facts about Rényi entropies and measure convergence that are used in Section 1.

Lemma A.3 (Continuity and uniform bound for finite-dimensional Rényi entropy)**.** *Let $p = (p_1, \ldots, p_d)$ be a probability vector, let $s := |\operatorname{supp} p|$, and use the standard endpoint definitions*

$$H_0(p) = \log s, \qquad H_1(p) = -\sum_{i:p_i>0} p_i \log p_i, \qquad H_\infty(p) = -\log \max_i p_i. \tag{A25}$$

Then $\alpha \mapsto H_\alpha(p)$ is continuous on the compactified interval $[0, +\infty]$, and

$$0 \le H_\alpha(p) \le \log s \le \log d \qquad (\alpha \in [0, +\infty]). \tag{A26}$$

Proof. For $\alpha \in (0, +\infty) \setminus \{1\}$, write

$$Z(\alpha) := \sum_{i:p_i>0} p_i^\alpha, \qquad H_\alpha(p) = \frac{\log Z(\alpha)}{1-\alpha}. \tag{A27}$$

The function Z is smooth and strictly positive. At $\alpha = 1$, Taylor's theorem gives

$$Z(1+h) = 1 + h \sum_i p_i \log p_i + o(h), \tag{A28}$$

whence

$$\lim_{h\to 0} \frac{\log Z(1+h)}{-h} = -\sum_i p_i \log p_i = H_1(p). \tag{A29}$$

At $\alpha = 0$, each positive term p_i^α converges to one, so $Z(\alpha) \to s$ and $H_\alpha(p) \to \log s$. At $+\infty$, put $m := \max_i p_i$. If q is the number of maximisers, then

$$m^\alpha \le Z(\alpha) \le sm^\alpha. \tag{A30}$$

Taking logarithms, dividing by $1 - \alpha < 0$, and letting $\alpha \to +\infty$ yields $H_\alpha(p) \to -\log m$.

For the bound, the lower inequality follows from $Z(\alpha) \ge 1$ when $0 \le \alpha < 1$ and $Z(\alpha) \le 1$ when $\alpha > 1$. For the upper inequality, Jensen's inequality gives

$$Z(\alpha) \le s^{1-\alpha} \quad (0 < \alpha < 1), \qquad Z(\alpha) \ge s^{1-\alpha} \quad (\alpha > 1). \tag{A31}$$

In both cases, division by $1-\alpha$ gives $H_\alpha(p) \le \log s$. The endpoint inequalities follow by continuity. ∎

Lemma A.4 (Concavity and Schur concavity in the sufficient-condition ranges)**.** *For every $\alpha \in [0, 1]$, H_α is concave on each finite probability simplex. For every $\alpha \in [0, +\infty]$, H_α is Schur concave. In particular, if D is doubly stochastic, then*

$$H_\alpha(Dp) \ge H_\alpha(p). \tag{A32}$$

Proof. For $0 < \alpha < 1$, the function $p \mapsto \sum_i p_i^\alpha$ is concave and positive, while the logarithm is increasing and concave. Therefore, for $0 \le \lambda \le 1$,

$$\begin{aligned} \log \sum_i (\lambda p_i + (1-\lambda) q_i)^\alpha &\ge \log \left(\lambda \sum_i p_i^\alpha + (1-\lambda) \sum_i q_i^\alpha \right) \\ &\ge \lambda \log \sum_i p_i^\alpha + (1-\lambda) \log \sum_i q_i^\alpha. \end{aligned} \tag{A33}$$

Division by $1-\alpha>0$ proves concavity. For $\alpha=1$, the Hessian of $-\sum_i p_i \log p_i$ on the interior is the diagonal matrix with entries $-1/p_i$, so the Shannon entropy is concave, and continuity extends the result to the boundary. The case $\alpha=0$ is proved directly in (57).

For Schur concavity, suppose that p majorises q. Karamata's inequality gives

$$\sum_i p_i^\alpha \le \sum_i q_i^\alpha \quad (0<\alpha<1), \qquad \sum_i p_i^\alpha \ge \sum_i q_i^\alpha \quad (\alpha>1). \tag{A34}$$

For $0<\alpha<1$, the denominator $1-\alpha$ is positive; for $\alpha>1$, it is negative. In both cases, (A34) implies $H_\alpha(p) \le H_\alpha(q)$. Shannon entropy is Schur concave by the same Karamata argument applied to the concave function $-x\log x$. If p majorises q, then $|\operatorname{supp} p| \le |\operatorname{supp} q|$ and $\max_i p_i \ge \max_i q_i$, proving the endpoint cases 0 and $+\infty$. Since Dp is majorised by p for every doubly stochastic D, (A32) follows. □

For clarity, we state the measure-theoretic tools in the exact forms in which they are used.

Lemma A.5 (Measure-theoretic tools)**.** *Let $(\Omega, \mathcal{F}, \nu)$ be a finite positive measure space.*

1. *If $f_n \to f$ ν-almost everywhere and $|f_n| \le g$ for some $g \in L^1(\nu)$, then*
$$\int f_n \,\mathrm{d}\nu \longrightarrow \int f \,\mathrm{d}\nu. \tag{A35}$$
2. *If $0 \le f_n \uparrow f$ pointwise, then*
$$\int f_n \,\mathrm{d}\nu \uparrow \int f \,\mathrm{d}\nu, \tag{A36}$$
where the value $+\infty$ is allowed.
3. *If F is absolutely integrable with respect to the product of two finite measures, then its iterated integrals exist and agree with its product integral.*

The first, second, and third statements are respectively the dominated convergence theorem, the monotone convergence theorem, and Fubini's theorem. For a finite signed measure μ, they are applied to its total-variation measure $|\mu|$.

Lemma A.6 (Common discrete approximation of the Rényi parameter)**.** *Let τ be a finite positive measure on $[0,+\infty]$, let T_n be a finite-valued Borel map such that $T_n(\alpha) \to \alpha$ for τ-almost every α, and let $\tau_n := (T_n)_\# \tau$. Then, for every finite probability vector p,*

$$\int H_\beta(p)\,\mathrm{d}\tau_n(\beta) = \int H_{T_n(\alpha)}(p)\,\mathrm{d}\tau(\alpha) \longrightarrow \int H_\alpha(p)\,\mathrm{d}\tau(\alpha). \tag{A37}$$

If the same maps T_n are used for all vectors, then the same sequence of functions may be used simultaneously at every point in a concavity, convexity, or quasi-convexity inequality.

Proof. The equality in (A37) is the defining property of a push-forward measure. By Lemma A.3, the integrand converges pointwise for τ-almost every α and is bounded by the constant $\log d$. Since τ is finite, this constant belongs to $L^1(\tau)$. The dominated convergence theorem therefore proves the limit.

For the final assertion, fix $\vec{x}, \vec{z}$ and $\lambda \in [0,1]$. A curvature inequality for the nth approximating function compares its values at the three points $\vec{x}$, $\vec{z}$, and $\lambda\vec{x} + (1-\lambda)\vec{z}$. The use of a single T_n, rather than three vector-dependent approximations, ensures that these are values of one and the same function. Pointwise convergence at the three points then permits passage to the limit. □

Lemma A.7 (Pointwise limits preserve the shape inequalities)**.** *Let C be a convex set and suppose that $f_n : C \to \mathbb{R}$ converges pointwise to f. If every f_n is concave, convex, or quasi-convex, then f has the same respective property.*

Proof. For concavity, take the limit in

$$f_n(\lambda x + (1-\lambda)y) \geq \lambda f_n(x) + (1-\lambda) f_n(y). \tag{A38}$$

The convex case is identical with the opposite inequality. For quasi-convexity, take the limit in

$$f_n(\lambda x + (1-\lambda)y) \leq \max\{f_n(x), f_n(y)\}, \tag{A39}$$

using continuity of the maximum of two real numbers. □

2. Endpoint-aware differentiation and signed-measure integration

We now prove the fixed-dimensional differential identities used throughout Section 2. Keeping this step at finite dimension is essential: it is here that the cancellation at $\alpha = 1$ is preserved.

Let

$$\vec{x}(\lambda) = \vec{x} + \lambda \vec{v} \tag{A40}$$

be an affine path in $\mathbb{R}^d_{>0}$ for $|\lambda| \leq \lambda_0$. For $\alpha \in (0, +\infty)$, define

$$\ell_\alpha(\lambda) := \frac{1}{\alpha} \log \sum_{i=1}^{d} x_i(\lambda)^\alpha, \qquad \pi_i^{(\alpha)}(\lambda) := \frac{x_i(\lambda)^\alpha}{\sum_j x_j(\lambda)^\alpha}, \qquad u_i(\lambda) := \frac{v_i}{x_i(\lambda)}. \tag{A41}$$

The vector $\pi^{(\alpha)}(\lambda)$ is the α-escort probability vector. For any coordinate arrays f and g, we specialize the preceding definitions as

$$\mathbb{E}_\alpha[f] := \sum_i \pi_i^{(\alpha)}(\lambda) f_i, \tag{A42}$$

$$\mathrm{Var}_\alpha(f) := \mathbb{E}_\alpha[f^2] - \big(\mathbb{E}_\alpha[f]\big)^2, \tag{A43}$$

$$\mathrm{Cov}_\alpha(f, g) := \mathbb{E}_\alpha[fg] - \mathbb{E}_\alpha[f]\mathbb{E}_\alpha[g]. \tag{A44}$$

Primes and superscripts (r) in this subsection denote derivatives in λ, while ∂_α denotes differentiation in the Rényi parameter.

Lemma A.8 (First and second derivatives of the logarithmic power mean)**.** *For every $\alpha \in (0, +\infty)$,*

$$\ell'_\alpha(\lambda) = \mathbb{E}_\alpha[u], \tag{A45}$$

$$\ell''_\alpha(\lambda) = -\mathbb{E}_\alpha[u^2] + \alpha \, \mathrm{Var}_\alpha(u). \tag{A46}$$

Equivalently,

$$\ell''_\alpha(\lambda) = (\alpha - 1)\mathbb{E}_\alpha[u^2] - \alpha\big(\mathbb{E}_\alpha[u]\big)^2. \tag{A47}$$

Proof. Let $S_\alpha(\lambda) := \sum_i x_i(\lambda)^\alpha$. Since $x_i' = v_i = x_i u_i$,

$$S_\alpha' = \alpha \sum_i x_i^{\alpha-1} v_i = \alpha \sum_i x_i^\alpha u_i = \alpha S_\alpha \mathbb{E}_\alpha[u]. \tag{A48}$$

Therefore

$$\ell_\alpha' = \frac{1}{\alpha}\frac{S_\alpha'}{S_\alpha} = \mathbb{E}_\alpha[u], \tag{A49}$$

which proves (A45).

The two elementary derivatives needed for the second derivative are

$$u_i' = -u_i^2 \tag{A50}$$

and

$$\begin{aligned}(\pi_i^{(\alpha)})' &= \alpha \pi_i^{(\alpha)} u_i - \pi_i^{(\alpha)} \frac{S_\alpha'}{S_\alpha} \\ &= \alpha \pi_i^{(\alpha)} \big(u_i - \mathbb{E}_\alpha[u]\big).\end{aligned} \tag{A51}$$

Differentiating (A45) and using (A50)–(A51) gives

$$\begin{aligned}\ell_\alpha'' &= \sum_i (\pi_i^{(\alpha)})' u_i + \sum_i \pi_i^{(\alpha)} u_i' \\ &= \alpha\left(\mathbb{E}_\alpha[u^2] - \big(\mathbb{E}_\alpha[u]\big)^2\right) - \mathbb{E}_\alpha[u^2],\end{aligned} \tag{A52}$$

which is (A46); rearrangement gives (A47). □

For $\alpha \neq 1$, the normalised Rényi entropy can be written without separating its cancelling terms:

$$\begin{aligned}H_\alpha(\widehat{x(\lambda)}) &= \frac{1}{1-\alpha}\left(\log \sum_i x_i(\lambda)^\alpha - \alpha \log \sum_i x_i(\lambda)\right) \\ &= \omega(\alpha)\big(\ell_\alpha(\lambda) - \ell_1(\lambda)\big),\end{aligned} \tag{A53}$$

where $\ell_1(\lambda) = \log \|\vec{x}(\lambda)\|_1$.

Lemma A.9 (Cancellation and continuity through the Shannon point)**.** *For $r = 0, 1, 2$, the map*

$$(\alpha, \lambda) \longmapsto \frac{\mathrm{d}^r}{\mathrm{d}\lambda^r} H_\alpha(\widehat{x(\lambda)}) \tag{A54}$$

extends continuously through $\alpha = 1$. More precisely, for α close to 1,

$$\begin{aligned}&\omega(\alpha)\left(\ell_\alpha^{(r)}(\lambda) - \ell_1^{(r)}(\lambda)\right) \\ &\qquad = -\alpha \int_0^1 \partial_\beta \ell_\beta^{(r)}(\lambda)\Big|_{\beta = 1 + s(\alpha-1)} \mathrm{d}s.\end{aligned} \tag{A55}$$

Proof. Because every coordinate $x_i(\lambda)$ is strictly positive on the compact interval, the finite sum

$$(\alpha, \lambda) \longmapsto \sum_i \exp\bigl(\alpha \log x_i(\lambda)\bigr) \tag{A56}$$

is smooth and strictly positive on a neighbourhood of $[1-\eta, 1+\eta] \times [-\lambda_0, \lambda_0]$ for sufficiently small $\eta > 0$. The factor $1/\alpha$ is smooth there. Hence $\ell_\alpha^{(r)}(\lambda)$ is continuously differentiable in α for $r = 0, 1, 2$.

The fundamental theorem of calculus in the α variable yields

$$\ell_\alpha^{(r)}(\lambda) - \ell_1^{(r)}(\lambda) = (\alpha - 1) \int_0^1 \partial_\beta \ell_\beta^{(r)}(\lambda)\Big|_{\beta = 1 + s(\alpha - 1)} \, \mathrm{d}s. \tag{A57}$$

Since $\omega(\alpha)(\alpha - 1) = -\alpha$, multiplication gives (A55). The right-hand side is jointly continuous at $\alpha = 1$. It therefore defines the continuous Shannon value of the rth λ derivative. Notice that no bound involving $1/|1-\alpha|$ has been introduced. □

The parameter endpoints 0 and $+\infty$ are also needed because the representing measure may have atoms there.

Lemma A.10 (Continuity of the differentiated entropy at the parameter endpoints)**.** *For the strictly positive affine path above, the first two λ derivatives of $H_\alpha(\widehat{x(\lambda)})$ extend continuously to $\alpha = 0$. They vanish at $\alpha = 0$.*

Assume, in addition, that there is a nonempty set I of coordinates, a positive C^2 function $m(\lambda)$, and a constant $\rho < 1$ such that, for all $|\lambda| \le \lambda_0$,

$$x_i(\lambda) = m(\lambda) \quad (i \in I), \qquad \max_{i \notin I} \frac{x_i(\lambda)}{m(\lambda)} \le \rho. \tag{A58}$$

Then the first two derivatives also extend continuously to $\alpha = +\infty$, with limiting entropy

$$H_\infty(\widehat{x(\lambda)}) = \ell_1(\lambda) - \log m(\lambda). \tag{A59}$$

Proof. For the endpoint 0, put $p_i(\lambda) := x_i(\lambda)/\|\vec{x}(\lambda)\|_1$. Compactness and strict positivity imply $p_i(\lambda) \ge m_0 > 0$. Hence

$$Z(\alpha, \lambda) := \sum_i \exp\bigl(\alpha \log p_i(\lambda)\bigr) \tag{A60}$$

is smooth jointly in (α, λ) on a neighbourhood of $[0, \eta] \times [-\lambda_0, \lambda_0]$. Since $1 - \alpha$ does not vanish there,

$$H_\alpha(p(\lambda)) = \frac{\log Z(\alpha, \lambda)}{1 - \alpha} \tag{A61}$$

is smooth up to $\alpha = 0$. At $\alpha = 0$, $Z(0, \lambda) = d$, independently of λ, so $H_0 = \log d$ and its first two λ derivatives vanish.

For $+\infty$, write $q := |I|$ and

$$r_i(\lambda) := \frac{x_i(\lambda)}{m(\lambda)} \quad (i \notin I), \qquad R_\alpha(\lambda) := \sum_{i \notin I} r_i(\lambda)^\alpha. \tag{A62}$$

Then

$$\ell_\alpha(\lambda) = \log m(\lambda) + \frac{1}{\alpha}\log\bigl(q + R_\alpha(\lambda)\bigr). \tag{A63}$$

On the compact λ interval, the functions r_i, r_i'/r_i, and their first derivatives are bounded, while $0 < r_i \le \rho$. Therefore, for constants independent of α and λ,

$$|R_\alpha(\lambda)| \le C\rho^\alpha, \tag{A64}$$

$$|R_\alpha'(\lambda)| \le C\alpha\rho^\alpha, \tag{A65}$$

$$|R_\alpha''(\lambda)| \le C\alpha^2\rho^\alpha. \tag{A66}$$

Since $q + R_\alpha \ge q$, differentiation of (A63) gives

$$\sup_{|\lambda|\le\lambda_0} \bigl|\ell_\alpha'(\lambda) - (\log m)'(\lambda)\bigr| \le C\rho^\alpha, \tag{A67}$$

$$\sup_{|\lambda|\le\lambda_0} \bigl|\ell_\alpha''(\lambda) - (\log m)''(\lambda)\bigr| \le C(1+\alpha)\rho^\alpha. \tag{A68}$$

At order zero, the additional term $(\log q)/\alpha$ tends to zero. Since $\omega(\alpha) \to -1$ and is bounded for $\alpha \ge 2$, insertion into the cancelled identity (A53) proves uniform convergence of the first two entropy derivatives to those of (A59). □

Every block path in Section 2 has a stable largest-coordinate block, so the hypothesis (A58) is satisfied.

Proposition A.11 (Differentiation under the Rényi-parameter integral at fixed dimension)**.** *Let μ be a finite signed measure on $[0, +\infty]$, and let $\vec{x}(\lambda)$ be a strictly positive affine path on $[-\lambda_0, \lambda_0]$ satisfying the stable-maximum condition (A58). Then*

$$F(\lambda) := \int_{[0,+\infty]} H_\alpha(\widehat{x(\lambda)})\,\mathrm{d}\mu(\alpha) \tag{A69}$$

is twice continuously differentiable and

$$F^{(r)}(\lambda) = \int_{[0,+\infty]} \frac{\mathrm{d}^r}{\mathrm{d}\lambda^r} H_\alpha(\widehat{x(\lambda)})\,\mathrm{d}\mu(\alpha), \qquad r = 1, 2. \tag{A70}$$

No weighted moment condition at $\alpha = 1$ is required.

Proof. By Lemmas A.9 and A.10, for $r = 0, 1, 2$ the map

$$f_r(\alpha, \lambda) := \frac{\mathrm{d}^r}{\mathrm{d}\lambda^r} H_\alpha(\widehat{x(\lambda)}) \tag{A71}$$

is continuous on the compact set $[0, +\infty] \times [-\lambda_0, \lambda_0]$. In particular, f_1 and f_2 are bounded and uniformly continuous.

Fix λ in the interior and take $h \ne 0$ small enough that $\lambda + sh$ remains in the interval for every $s \in [0, 1]$. The fundamental theorem of calculus gives, pointwise in α,

$$\frac{f_0(\alpha, \lambda + h) - f_0(\alpha, \lambda)}{h} = \int_0^1 f_1(\alpha, \lambda + sh)\,\mathrm{d}s. \tag{A72}$$

The right-hand side is bounded by $\sup|f_1|$, which is integrable with respect to the finite measure $|\mu|$. Fubini's theorem therefore permits integration in either order. Subtracting the proposed derivative gives

$$\begin{aligned} &\frac{F(\lambda+h)-F(\lambda)}{h} - \int f_1(\alpha,\lambda)\,\mathrm{d}\mu(\alpha) \\ &\quad = \int\int_0^1 \left[f_1(\alpha,\lambda+sh) - f_1(\alpha,\lambda)\right]\,\mathrm{d}s\,\mathrm{d}\mu(\alpha). \end{aligned} \tag{A73}$$

Its absolute value is at most

$$|\mu|([0,+\infty]) \sup_{\substack{\alpha\in[0,+\infty]\\ s\in[0,1]}} \left|f_1(\alpha,\lambda+sh) - f_1(\alpha,\lambda)\right|, \tag{A74}$$

which tends to zero by uniform continuity of f_1. This proves the case $r=1$. Applying the same argument with f_1 in place of f_0 and f_2 in place of f_1 proves the case $r=2$. Continuity of the resulting derivatives follows from the same uniform-continuity estimate. □

3. Uniform two-block and three-block localization

We next prove the high-dimensional formulas used in Section 2. The exact finite-dimensional derivatives are always taken first, according to Proposition A.11. Only then do we send d to infinity.

a. *General block estimates*

Consider a vector divided into finitely many coordinate blocks. At $\lambda=0$, assume that block j has a common relative velocity u_j and aggregate power weight

$$\Pi_{j,d}^{(\alpha)} := \frac{\sum_{i\in j} x_i^\alpha}{\sum_i x_i^\alpha}. \tag{A75}$$

These numbers are probability weights on the blocks. Here $\ell_{\alpha,d}$ means ℓ_α for the d-dimensional block path, and unsubscripted $\mathbb{E}$, Var, and Cov in this subsection use the weights $\Pi_d^{(\alpha)}$. Then Lemma A.8 becomes

$$\ell'_{\alpha,d}(0) = \sum_j \Pi_{j,d}^{(\alpha)} u_j, \tag{A76}$$

$$\ell''_{\alpha,d}(0) = -\sum_j \Pi_{j,d}^{(\alpha)} u_j^2 + \alpha\,\mathrm{Var}_{\Pi_d^{(\alpha)}}(u). \tag{A77}$$

Lemma A.12 (Dominant-block remainder)**.** *Let $M := \max_j |u_j|$. If block j_* has power weight at least $1-\delta$, then*

$$\left|\ell'_{\alpha,d}(0) - u_{j_*}\right| \le 2M\delta, \tag{A78}$$

$$\left|\ell''_{\alpha,d}(0) + u_{j_*}^2\right| \le 2M^2(1+2\alpha)\delta. \tag{A79}$$

Proof. Since the total off-block weight is at most δ,

$$|\mathbb{E}[u]-u_{j_*}| \le \sum_{j\neq j_*} \Pi_j |u_j-u_{j_*}| \le 2M\delta, \tag{A80}$$

$$\left|\mathbb{E}[u^2]-u_{j_*}^2\right| \le \sum_{j\neq j_*} \Pi_j |u_j^2-u_{j_*}^2| \le 2M^2\delta. \tag{A81}$$

Moreover,

$$\mathrm{Var}(u) \le \mathbb{E}[(u-u_{j_*})^2] \le 4M^2\delta. \tag{A82}$$

Substitution into (A77) proves (A79). □

The estimates near $\alpha = 1$ use differentiation of power weights in the Rényi parameter. If f is constant on each block, then

$$\partial_\alpha \mathbb{E}_\alpha[f] = \mathrm{Cov}_\alpha(f, \log x). \tag{A83}$$

Indeed,

$$\partial_\alpha \pi_i^{(\alpha)} = \pi_i^{(\alpha)} \left(\log x_i - \mathbb{E}_\alpha[\log x]\right), \tag{A84}$$

and summation against f_i yields (A83).

If f and g are constant on the dominant block, with values f_* and g_* there, then

$$\begin{aligned} |\operatorname{Cov}(f,g)| &= |\mathbb{E}[(f-f_*)(g-g_*)] - \mathbb{E}[f-f_*]\mathbb{E}[g-g_*]| \\ &\le 2\|f-f_*\|_\infty \|g-g_*\|_\infty \delta. \end{aligned} \tag{A85}$$

This estimate is the source of the factor $(\log d)d^{-g}$ below.

b. *The two-block localiser*

We prove the formulas (73)–(77). At $\lambda = 0$, write

$$n_{0,d} = \theta_{0,d} d^R, \qquad n_{1,d} = \theta_{1,d} d^{R-r}, \qquad 1 \le \theta_{j,d} \le 2. \tag{A86}$$

The power sum of (70) is

$$\sum_i x_i(\lambda)^\alpha = d^R\left[\theta_{0,d}(1+u_0\lambda)^\alpha + \theta_{1,d} d^{\alpha-r}(1+u_1\lambda)^\alpha\right]. \tag{A87}$$

Thus the exponent gap between the two blocks is $|\alpha - r|$. On every compact set $K \subset [0, r)$, the off-dominant weight satisfies

$$\Pi_{1,d}^{(\alpha)} \le C_K d^{-g_K}, \qquad g_K := \inf_{\alpha\in K}(r-\alpha) > 0, \tag{A88}$$

and on every compact $K \subset (r, +\infty)$,

$$\Pi_{0,d}^{(\alpha)} \le C_K d^{-g_K}, \qquad g_K := \inf_{\alpha\in K}(\alpha - r) > 0. \tag{A89}$$

Consequently, Lemma A.12 gives

$$\ell'_{\alpha,d}(0) \longrightarrow u_0, \qquad \ell''_{\alpha,d}(0) \longrightarrow -u_0^2 \qquad (\alpha < r), \tag{A90}$$

$$\ell'_{\alpha,d}(0) \longrightarrow u_1, \qquad \ell''_{\alpha,d}(0) \longrightarrow -u_1^2 \qquad (\alpha > r). \tag{A91}$$

Let $k = 0$ if $r > 1$ and $k = 1$ if $r < 1$; this is the block which dominates at $\alpha = 1$. Combining (A90)–(A91) with the cancelled identity (A53) gives, for $j \neq k$ and for α in the cell where block j dominates,

$$H'_\alpha(0) \longrightarrow \omega(\alpha)(u_j - u_k), \tag{A92}$$

$$H''_\alpha(0) \longrightarrow \omega(\alpha)(u_k^2 - u_j^2), \tag{A93}$$

whereas both limits are zero in the cell containing 1.

We now justify integration of these limits. Choose $\eta > 0$ such that $[1-\eta, 1+\eta]$ is contained in the cell containing 1. The exponent gap on this interval has a positive lower bound g. At $\lambda = 0$, the range of $\log x_i$ is at most $\log d$, and (A85) gives

$$|\partial_\alpha \ell'_{\alpha,d}(0)| = |\operatorname{Cov}_\alpha(u, \log x)| \leq CM(\log d)d^{-g}. \tag{A94}$$

Using (A47), differentiation in α gives the exact identity

$$\begin{aligned}\partial_\alpha \ell''_{\alpha,d}(0) = {} & \operatorname{Var}_\alpha(u) + (\alpha - 1)\operatorname{Cov}_\alpha(u^2, \log x) \\ & - 2\alpha \mathbb{E}_\alpha[u] \operatorname{Cov}_\alpha(u, \log x).\end{aligned} \tag{A95}$$

The variance is $O(M^2 d^{-g})$ by (A82), and both covariances are $O(M^2(\log d)d^{-g})$. Since α stays in a compact neighbourhood of 1,

$$|\partial_\alpha \ell''_{\alpha,d}(0)| \leq CM^2(\log d)d^{-g}. \tag{A96}$$

Applying the cancellation formula (A55) with $r = 1, 2$ yields

$$\sup_{|\alpha-1|\leq\eta} |H_\alpha^{(s)}(0)| \leq C_s(\log d)d^{-g}, \qquad s = 1, 2. \tag{A97}$$

On any fixed compact subset separated from 1, the elementary estimates

$$|\ell'_{\alpha,d}(0)| \leq M, \qquad |\ell''_{\alpha,d}(0)| \leq (1 + 4\alpha)M^2 \tag{A98}$$

combine with boundedness of ω to give a constant independent of d. This covers a neighbourhood of the crossing point r, because $r \neq 1$.

For the unbounded tail, take $\alpha \geq \max\{r + 1, 2\}$. Directly from (A87),

$$\Pi_{0,d}^{(\alpha)} \leq Cd^{r-\alpha} \leq C2^{-(\alpha - r)}. \tag{A99}$$

The dominant-block estimate then bounds the apparently growing quantity $\alpha \operatorname{Var}(u)$ by

$$CM^2 \alpha 2^{-(\alpha - r)}, \tag{A100}$$

which is uniformly bounded. Since $|\omega(\alpha)| \leq 2$ for $\alpha \geq 2$, both $H'_\alpha(0)$ and $H''_\alpha(0)$ are bounded by one constant, independently of d and α.

Together, (A97), the compact-set bound, and the tail bound provide a constant C_M such that

$$|H'_\alpha(0)| + |H''_\alpha(0)| \leq C_M \quad \text{for all } d \geq 2,\ \alpha \in [0, +\infty]. \tag{A101}$$

The constant is integrable against the finite measure $|\mu|$. The only point where the pointwise limit can fail is r, and r was chosen $|\mu|$-null. Dominated convergence therefore permits the $d \to \infty$ limit to pass through the parameter integral.

If $r > 1$, the off-Shannon cell is $(r, +\infty]$ and its coefficient is A_r. Since

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} \log \|\vec{x}_r^{(d)}(\lambda)\|_1 \Big|_0 = \ell'_{1,d}(0) \longrightarrow u_0, \qquad \ell''_{1,d}(0) \longrightarrow -u_0^2, \tag{A102}$$

we obtain

$$\Lambda'_d(0) \longrightarrow u_0 + A_r(u_1 - u_0) = (1 - A_r)u_0 + A_r u_1, \tag{A103}$$

$$\Lambda''_d(0) \longrightarrow -u_0^2 + A_r(u_0^2 - u_1^2) = -(1 - A_r)u_0^2 - A_r u_1^2. \tag{A104}$$

This proves (73)–(74). If $r < 1$, the off-Shannon cell is $[0, r)$ and the same calculation with B_r and u_1 as the norm-dominant velocity gives

$$\Lambda'_d(0) \longrightarrow B_r u_0 + (1 - B_r)u_1, \tag{A105}$$

$$\Lambda''_d(0) \longrightarrow -B_r u_0^2 - (1 - B_r)u_1^2, \tag{A106}$$

which proves (76)–(77). Every estimate depends on the velocities only through M, so the convergence is locally uniform in (u_0, u_1).

For the tropical function G_μ, the norm term is absent. Thus the corresponding limits are obtained from the preceding formulas by subtracting the limiting norm derivatives.

c. *The three-block localiser*

We now prove (82)–(83). Write

$$\lceil d^R \rceil = \theta_{0,d} d^R, \quad \lceil d^{R-a} \rceil = \theta_{1,d} d^{R-a}, \quad \lceil d^{R-a-b} \rceil = \theta_{2,d} d^{R-a-b}, \qquad 1 \le \theta_{j,d} \le 2. \tag{A107}$$

The power sum of (78) is

$$\sum_i x_i(\lambda)^\alpha = d^R \big[\theta_{0,d}(1 + u_0\lambda)^\alpha + \theta_{1,d} d^{\alpha - a}(1 + u_1\lambda)^\alpha + \theta_{2,d} d^{2\alpha - a - b}(1 + u_2\lambda)^\alpha\big]. \tag{A108}$$

The pairwise differences of the exponents in (79) are

$$q_1 - q_0 = \alpha - a, \qquad q_2 - q_1 = \alpha - b, \qquad q_2 - q_0 = 2\alpha - a - b. \tag{A109}$$

It follows directly that block 0 is uniquely dominant on $J_0 = [0, a)$, block 1 on $J_1 = (a, b)$, and block 2 on $J_2 = (b, +\infty]$.

On a compact set K contained in one open cell, let

$$g_K := \min_{\alpha \in K} \Big(q_{j_*}(\alpha) - \max_{j \ne j_*} q_j(\alpha) \Big) > 0. \tag{A110}$$

Equation (A108) then implies that the total off-dominant power weight is at most

$$\delta_{\alpha,d} \le C_K d^{-g_K}. \tag{A111}$$

Lemma A.12 therefore gives, pointwise away from a and b,

$$\ell'_{\alpha,d}(0) \longrightarrow u_j, \qquad \ell''_{\alpha,d}(0) \longrightarrow -u_j^2 \quad (\alpha \in J_j). \tag{A112}$$

Let k be the unique cell containing 1. The norm derivatives satisfy

$$\ell'_{1,d}(0) \longrightarrow u_k, \qquad \ell''_{1,d}(0) \longrightarrow -u_k^2. \tag{A113}$$

Inserting these limits into (A53) gives

$$H'_\alpha(0) \longrightarrow \omega(\alpha)(u_j - u_k), \tag{A114}$$

$$H''_\alpha(0) \longrightarrow \omega(\alpha)(u_k^2 - u_j^2) \tag{A115}$$

for $\alpha \in J_j$, $j \neq k$, and zero limits in J_k.

The domination argument is the same in structure as for two blocks, but we spell out the changes. Choose $\eta > 0$ such that $[1-\eta, 1+\eta] \subset J_k$. The exponent gap from the dominant block is then bounded below by some $g > 0$. At $\lambda = 0$, the coordinate logarithms are 0, $\log d$, and $2\log d$, up to no additional d-dependent term. Equations (A83), (A85), and (A95) give

$$|\partial_\alpha \ell'_{\alpha,d}(0)| \le CM(\log d)d^{-g}, \tag{A116}$$

$$|\partial_\alpha \ell''_{\alpha,d}(0)| \le CM^2(\log d)d^{-g}. \tag{A117}$$

The cancellation formula yields

$$\sup_{|\alpha-1|\le\eta} |H_\alpha^{(s)}(0)| \le C_s(\log d)d^{-g}, \qquad s = 1, 2. \tag{A118}$$

On compact sets separated from 1, the rough bounds (A98) apply and ω is bounded. This includes fixed neighbourhoods of a and b.

For the unbounded tail, take $\alpha \ge \max\{b+1, 2\}$. Dividing the first two terms of (A108) by the third gives

$$\frac{\theta_{1,d} d^{\alpha-a}}{\theta_{2,d} d^{2\alpha-a-b}} \le C d^{b-\alpha} \le C 2^{-(\alpha-b)}, \tag{A119}$$

$$\frac{\theta_{0,d}}{\theta_{2,d} d^{2\alpha-a-b}} \le C d^{-(2\alpha-a-b)} \le C 2^{-(\alpha-b)}. \tag{A120}$$

Thus

$$1 - \Pi_{2,d}^{(\alpha)} \le C 2^{-(\alpha-b)}. \tag{A121}$$

As before, $\alpha \operatorname{Var}(u)$ is bounded by a constant multiple of $\alpha 2^{-(\alpha-b)}$, and $|\omega| \le 2$ on this tail. Hence the two differentiated entropy functions admit a constant dominator independent of both d and α.

Since a and b are $|\mu|$-null, dominated convergence gives

$$\lim_{d\to\infty} \int H'_\alpha(0)\, \mathrm{d}\mu(\alpha) = \sum_{j\neq k} A_j (u_j - u_k), \tag{A122}$$

$$\lim_{d\to\infty} \int H''_\alpha(0)\, \mathrm{d}\mu(\alpha) = \sum_{j\neq k} A_j (u_k^2 - u_j^2). \tag{A123}$$

Adding the norm limits (A113) yields

$$\Lambda_d'(0) \longrightarrow \left(1 - \sum_{j\neq k} A_j\right) u_k + \sum_{j\neq k} A_j u_j, \tag{A124}$$

$$\Lambda_d''(0) \longrightarrow -\left(1 - \sum_{j\neq k} A_j\right) u_k^2 - \sum_{j\neq k} A_j u_j^2, \tag{A125}$$

which are (82)–(83). Omitting the norm term, define

$$G_d(\lambda) := G_\mu\big(\vec{x}_{a,b}^{(d)}(\lambda)\big). \tag{A126}$$

Then

$$G_d'(0) \longrightarrow -\left(\sum_{j\neq k} A_j\right) u_k + \sum_{j\neq k} A_j u_j, \tag{A127}$$

$$G_d''(0) \longrightarrow \left(\sum_{j\neq k} A_j\right) u_k^2 - \sum_{j\neq k} A_j u_j^2, \tag{A128}$$

which are (128)–(129). The constants in all bounds depend on the velocities only through M, so the convergence is locally uniform.

The preceding proof establishes the precise order

$$\boxed{\text{differentiate at fixed finite } d \;\longrightarrow\; \text{integrate in } \alpha \;\longrightarrow\; d \to +\infty.} \tag{A129}$$

No derivative of the limiting piecewise-affine exponent envelope is taken.

4. The separate perturbation at the Shannon point

The block localisers of Appendix A 3 deliberately suppress the entire cell containing $\alpha = 1$. They therefore cannot distinguish an atom at 1 from nonatomic mass near 1. The perturbation (85) is designed to make the Shannon first derivative converge to a prescribed finite number, while the derivatives at every fixed $\alpha \neq 1$ below the second crossing converge to zero.

Put

$$L_d := \log d, \qquad \theta_{0,d} := \frac{n_{0,d}}{d^R}, \qquad \theta_{1,d} := \frac{n_{1,d}}{d^{R-1}}, \qquad \theta_{2,d} := \frac{n_{2,d}}{d^{R-1-c}}. \tag{A130}$$

Also set

$$\varepsilon_d := \max_{j=0,1,2} |\theta_{j,d} - 1|, \qquad s_d := \theta_{2,d} d^{1-c}, \qquad M_d := q\theta_{0,d} + p\theta_{1,d}. \tag{A131}$$

The ceiling bounds give

$$\theta_{j,d} \longrightarrow 1, \qquad \varepsilon_d L_d \longrightarrow 0, \qquad s_d L_d \longrightarrow 0. \tag{A132}$$

Keeping the ceiling factors is important: it makes this calculation apply to the exact finite vector used in Lean. Its total mass is

$$\|\vec{x}_{\mathrm{Sh}}^{(d)}(\lambda)\|_1 = d^R \left[M_d + s_d + \lambda\left(\frac{h(\theta_{0,d} - \theta_{1,d})}{L_d} + s_d v\right)\right], \tag{A133}$$

so, writing

$$a_d := \frac{h(\theta_{0,d} - \theta_{1,d})}{L_d} + s_d v, \tag{A134}$$

we have

$$\ell'_{1,d}(0) = \frac{a_d}{M_d + s_d}, \tag{A135}$$

$$\ell''_{1,d}(0) = -\left(\frac{a_d}{M_d + s_d}\right)^2. \tag{A136}$$

Both converge to zero, locally uniformly in (h, v).

We next calculate the Shannon derivative exactly enough to identify its limit. For any strictly positive affine path, let

$$p_i(\lambda) := \frac{x_i(\lambda)}{\sum_j x_j(\lambda)}, \qquad w_i(\lambda) := \frac{x'_i(\lambda)}{x_i(\lambda)}. \tag{A137}$$

For a coordinate array f, set

$$\mathbb{E}_{p(\lambda)}[f] := \sum_i p_i(\lambda) f_i, \tag{A138}$$

$$\mathrm{Var}_{p(\lambda)}(f) := \mathbb{E}_{p(\lambda)}[f^2] - \big(\mathbb{E}_{p(\lambda)}[f]\big)^2, \tag{A139}$$

$$\mathrm{Cov}_{p(\lambda)}(f, g) := \mathbb{E}_{p(\lambda)}[fg] - \mathbb{E}_{p(\lambda)}[f]\mathbb{E}_{p(\lambda)}[g]. \tag{A140}$$

We abbreviate these subscripts to p when λ is clear, and write $(\log x)_i := \log x_i$. Thus expectation is the weighted average, variance measures the squared fluctuation about that average, and covariance measures the joint fluctuation of two arrays. Then

$$p'_i = p_i\big(w_i - \mathbb{E}_p[w]\big), \qquad w'_i = -w_i^2. \tag{A141}$$

Lemma A.13 (First two derivatives of Shannon entropy along an affine path)**.** *With the notation above,*

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} H_1(p(\lambda)) = -\mathrm{Cov}_{p(\lambda)}\big(w(\lambda), \log x(\lambda)\big), \tag{A142}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}\lambda^2} H_1(p(\lambda)) = -\mathrm{Var}_{p(\lambda)}(w(\lambda)) + 2\mathbb{E}_{p(\lambda)}[w(\lambda)]\,\mathrm{Cov}_{p(\lambda)}\big(w(\lambda), \log x(\lambda)\big). \tag{A143}$$

Proof. Since $\log p_i = \log x_i - \log \sum_j x_j$, covariance with $\log p$ equals covariance with $\log x$. Differentiating

$$H_1(p) = -\sum_i p_i \log p_i \tag{A144}$$

and using $\sum_i p'_i = 0$ gives

$$\begin{aligned} H'_1 &= -\sum_i p'_i \log p_i = -\sum_i p_i\big(w_i - \mathbb{E}_p[w]\big) \log p_i \\ &= -\mathrm{Cov}_p(w, \log p) = -\mathrm{Cov}_p(w, \log x), \end{aligned} \tag{A145}$$

which proves (A142).

For any differentiable observable $f_i(\lambda)$,

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathbb{E}_p[f] = \sum_i p_i' f_i + \sum_i p_i f_i'$$
$$= \mathrm{Cov}_p(f, w) + \mathbb{E}_p[f']. \tag{A146}$$

Applying this identity to the covariance in (A142), using $(\log x_i)' = w_i$ and $w_i' = -w_i^2$, gives

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\,\mathrm{Cov}_p(w, \log x) = \mathrm{Var}_p(w) - 2\mathbb{E}_p[w]\,\mathrm{Cov}_p(w, \log x). \tag{A147}$$

Negating this identity proves (A143). □

At $\lambda = 0$, the relative velocities in the three blocks are

$$w_{0,d} = \frac{h}{qL_d}, \qquad w_{1,d} = -\frac{h}{pL_d}, \qquad w_{2,d} = v. \tag{A148}$$

The normalised masses of the three blocks are

$$P_{0,d} = \frac{q\theta_{0,d}}{M_d + s_d}, \qquad P_{1,d} = \frac{p\theta_{1,d}}{M_d + s_d}, \qquad P_{2,d} = \frac{s_d}{M_d + s_d}. \tag{A149}$$

For the ideal first-two-block comparison in which the rounding factors are replaced by 1, the mean velocity is zero and direct substitution into (A142) gives

$$H_1'(0) = -\left[q\frac{h}{qL_d}\log q + p\left(-\frac{h}{pL_d}\right)(L_d + \log p)\right]$$
$$= h + \frac{h}{L_d}\log\frac{p}{q}. \tag{A150}$$

Likewise, (A143) reduces to the negative variance,

$$H_1''(0) = -\frac{h^2}{L_d^2}\left(\frac{1}{p} + \frac{1}{q}\right). \tag{A151}$$

The ceiling perturbation changes each of the first two normalized masses by $O(\varepsilon_d)$, while the third block has normalized mass $O(s_d)$. Its relative velocity is bounded on compact (h, v) sets, and its logarithmic coordinate differs from those of the first two blocks by $O(L_d)$. Equations (A142)–(A143) therefore give the uniform error estimates

$$H_1'(0) = h + \frac{h}{L_d}\log\frac{p}{q} + O\big((\varepsilon_d + s_d)L_d\big), \tag{A152}$$

$$H_1''(0) = -\frac{h^2}{L_d^2}\left(\frac{1}{p} + \frac{1}{q}\right) + O\big((\varepsilon_d + s_d)L_d\big). \tag{A153}$$

In view of (A132),

$$H_1'(0) \longrightarrow h, \qquad H_1''(0) \longrightarrow 0. \tag{A154}$$

The power-sum exponents are those in (86). Therefore, for every fixed $\alpha \neq 1, c$, the dominant-block estimate gives

$$H'_\alpha(0) \longrightarrow 0, \qquad H''_\alpha(0) \longrightarrow 0 \qquad (\alpha < c), \tag{A155}$$

$$H'_\alpha(0) \longrightarrow \omega(\alpha)v, \qquad H''_\alpha(0) \longrightarrow -\omega(\alpha)v^2 \qquad (\alpha > c). \tag{A156}$$

We now prove the uniform bounds needed for dominated convergence.

Choose $0 < \eta < (c-1)/2$. On $|\alpha - 1| \leq \eta$, the third block's power weight is at most $Cd^{-\kappa}$ for some $\kappa > 0$, because its exponent is uniformly below the maximum of the first two exponents. The first two velocities differ by $O(L_d^{-1})$, while their logarithmic coordinates differ by $L_d + O(1)$. Let

$$E_{\alpha,d} := \mathbb{E}_{\pi^{(\alpha)}}[w]. \tag{A157}$$

Using (A83), we obtain

$$|\partial_\alpha E_{\alpha,d}| = |\operatorname{Cov}_\alpha(w, \log x)| \leq C\big(1 + L_d d^{-\kappa}\big). \tag{A158}$$

The cancellation identity (A55) with $r = 1$ therefore implies

$$\sup_{|\alpha-1|\leq\eta} |H'_\alpha(0)| \leq C. \tag{A159}$$

For the second derivative, put

$$F_{\alpha,d} := \ell''_{\alpha,d}(0) = (\alpha - 1)\mathbb{E}_\alpha[w^2] - \alpha E^2_{\alpha,d}. \tag{A160}$$

The exact derivative (A95) gives

$$\begin{aligned}\partial_\alpha F_{\alpha,d} = \ &\operatorname{Var}_\alpha(w) + (\alpha - 1)\operatorname{Cov}_\alpha(w^2, \log x)\\ &- 2\alpha E_{\alpha,d} \operatorname{Cov}_\alpha(w, \log x).\end{aligned} \tag{A161}$$

For the first two blocks, the three terms on the right are respectively $O(L_d^{-2})$, $O(L_d^{-1})$, and $O(L_d^{-1})$. Terms involving the third block are $O(L_d d^{-\kappa})$. Thus

$$\sup_{|\alpha-1|\leq\eta} |\partial_\alpha F_{\alpha,d}| \leq C\left(L_d^{-1} + L_d d^{-\kappa}\right). \tag{A162}$$

Applying (A55) with $r = 2$ gives

$$\sup_{|\alpha-1|\leq\eta} |H''_\alpha(0)| \leq C\left(L_d^{-1} + L_d d^{-\kappa}\right) \longrightarrow 0. \tag{A163}$$

This estimate is the precise reason why the nonatomic measure in a whole neighbourhood of 1 contributes nothing to the limiting second derivative.

On compact sets separated from 1, the velocities (A148) are uniformly bounded and the rough estimates (A98) apply. Since ω is bounded there, this gives a constant dominator, including on a neighbourhood of the second crossing point c.

For $\alpha \geq \max\{c + 1, 2\}$, comparison with the third power-sum block gives

$$\begin{aligned}1 - \Pi^{(\alpha)}_{2,d} &\leq C\left(d^{c-\alpha} + d^{c+1-2\alpha}\right)\\ &\leq C2^{-(\alpha-c)}.\end{aligned} \tag{A164}$$

The factor α in the variance term is consequently bounded by a multiple of $\alpha 2^{-(\alpha-c)}$. Since $|\omega(\alpha)| \le 2$ on this tail, the first two differentiated entropies are bounded by a constant independent of d and α.

We have therefore constructed one constant $C_{h,v}$, locally uniform in (h,v), such that

$$|H'_\alpha(0)| + |H''_\alpha(0)| \le C_{h,v} \quad \text{for all } \alpha \in [0,+\infty],\ d \ge 2. \tag{A165}$$

The only point where the pointwise limit may fail is c, which was chosen $|\mu|$-null. Dominated convergence, together with (A154), (A155), and (A156), gives

$$\int H'_\alpha(0)\,\mathrm{d}\mu(\alpha) \longrightarrow mh + A_c v, \tag{A166}$$

$$\int H''_\alpha(0)\,\mathrm{d}\mu(\alpha) \longrightarrow -A_c v^2. \tag{A167}$$

Finally, the norm derivatives (A135)–(A136) vanish. This proves (88)–(89), as well as the identical limits for the tropical function G_μ. All estimates are uniform when (h,v) ranges over a compact set.

5. Exact stationarity and the tropical second-order argument

The negative tropical homomorphism is controlled by quasi-convexity. It is important that quasi-convexity does not imply convexity and does not impose a sign on the second derivative at a nonstationary point.

Lemma A.14 (Second-order condition for a one-dimensional quasi-convex function)**.** *Let g be twice differentiable in a neighbourhood of* 0*. If g is quasi-convex and $g'(0) = 0$, then*

$$g''(0) \ge 0. \tag{A168}$$

Proof. Suppose, to the contrary, that $g''(0) < 0$. Taylor's theorem gives

$$g(\pm h) = g(0) + \frac{h^2}{2} g''(0) + o(h^2), \tag{A169}$$

because the linear term vanishes. Hence $g(h) < g(0)$ and $g(-h) < g(0)$ for all sufficiently small $h > 0$. Quasi-convexity at the midpoint gives

$$g(0) = g\left(\frac{-h+h}{2}\right) \le \max\{g(-h), g(h)\} < g(0), \tag{A170}$$

a contradiction. □

The limiting counterexample velocities in Section 2 make the limiting first derivative zero. This alone is not sufficient: the derivative must be exactly zero at the same finite d at which Lemma A.14 is applied. The following elementary correction supplies exact stationarity.

In the next lemma, $g_{d,z}$ denotes the one-variable restriction obtained by using the block-velocity vector z at dimension d; its primes are λ-derivatives at zero. The dot product is $c_d \cdot z := \sum_j (c_d)_j z_j$.

Lemma A.15 (Exact finite-dimensional stationarity correction)**.** *Let $z \in \mathbb{R}^m$ denote the vector of block velocities. Suppose that, for every d, the first derivative is exactly linear in z,*

$$g'_{d,z}(0) = c_d \cdot z, \tag{A171}$$

where $c_d \to c$. Suppose also that

$$g''_{d,z}(0) \longrightarrow Q(z) \tag{A172}$$

locally uniformly in z. Let z satisfy $c \cdot z = 0$ and choose an index k with $c_k \neq 0$. Define

$$(z_d)_j := z_j \quad (j \neq k), \qquad (z_d)_k := z_k - \frac{c_d \cdot z}{(c_d)_k}. \tag{A173}$$

Then, for every sufficiently large d,

$$g'_{d,z_d}(0) = 0, \qquad z_d \longrightarrow z, \qquad g''_{d,z_d}(0) \longrightarrow Q(z). \tag{A174}$$

Proof. Since $(c_d)_k \to c_k \neq 0$, the denominator in (A173) is nonzero for all sufficiently large d. By construction,

$$c_d \cdot z_d = c_d \cdot z + (c_d)_k \left(-\frac{c_d \cdot z}{(c_d)_k} \right) = 0, \tag{A175}$$

which proves exact stationarity. Moreover, $c_d \cdot z \to c \cdot z = 0$, so the correction tends to zero and $z_d \to z$. Local uniform convergence of the second derivatives then gives the final statement. □

For all block paths used in Section 2, the base point at $\lambda = 0$ is independent of the velocity vector. Directional differentiation is therefore exactly linear in the velocities, as required in (A171). The locally uniform convergence was proved in Appendices A 3 and A 4. Since z_d remains bounded, the corrected affine paths remain strictly positive on a common neighbourhood of zero.

At exact stationarity, the exponential does not introduce an additional second-order term:

$$\frac{(e^{g_d})''(0)}{e^{g_d(0)}} = g''_d(0) + (g'_d(0))^2 = g''_d(0). \tag{A176}$$

This simplification is used only after (A174) has been imposed.

6. Finite semiring witnesses and the channel lift

The high-dimensional calculations produce a strict limiting sign. We conclude by explaining why this yields genuine finite-dimensional counterexamples, rather than merely formal tangent obstructions.

Let $f_d(\lambda)$ be the restriction of the relevant one-column function to one of the affine paths. If

$$f''_d(0) > 0, \tag{A177}$$

then Taylor's theorem gives

$$\frac{f_d(\varepsilon) + f_d(-\varepsilon)}{2} - f_d(0) = \frac{\varepsilon^2}{2} f''_d(0) + o(\varepsilon^2) > 0 \tag{A178}$$

for every sufficiently small $\varepsilon > 0$. Since

$$\vec{x}^{(d)}(0) = \frac{1}{2}\vec{x}^{(d)}(\varepsilon) + \frac{1}{2}\vec{x}^{(d)}(-\varepsilon), \tag{A179}$$

this is a direct violation of concavity. Similarly, $f''_d(0) < 0$ gives a direct violation of convexity.

In the tropical case, let g_d be the logarithmic tropical function after the exact stationarity correction of Lemma A.15. If $g_d''(0) < 0$, then

$$g_d(\varepsilon) < g_d(0), \qquad g_d(-\varepsilon) < g_d(0) \tag{A180}$$

for every sufficiently small $\varepsilon > 0$. Hence

$$g_d(0) > \max\{g_d(-\varepsilon), g_d(\varepsilon)\}, \tag{A181}$$

which is exactly the midpoint violation of quasi-convexity.

Finally, the endpoints are admissible semiring elements. Every base vector used in Section 2 is strictly positive. For any fixed finite d and bounded velocity vector, there exists $\varepsilon_d > 0$ such that all coordinates of $\vec{x}^{(d)}(\lambda)$ are positive for $|\lambda| < \varepsilon_d$. Thus the two endpoints in (A179) are positive one-column matrices in the original semiring. The notation col denotes vertical concatenation inside that one column; no semiring direct sum is involved.

[1] R. Rubboli, E. Haapasalo, and M. Tomamichel. *"A Complete Characterisation of Conditional Entropies"*. Manuscript, (2026). Available online: https://arxiv.org/abs/2601.23213. The equation and theorem numbers cited by the companion papers refer to the Ultra Lean revision.

[2] R. Rubboli, E. Haapasalo, and M. Tomamichel. *"Lean Formalization of the Parameter Classification for Conditional Entropies"*. Standalone formalization note, (2026). Dependency graph, trust audit, and statement correspondence.

[3] R. Rubboli, E. Haapasalo, and M. Tomamichel. *"Sufficient and Necessary Conditions for the Parameters of Conditional Entropies"*. Companion note, (2026). Concise presentation of the exact parameter classification.